

CCNA SPEEDRUN

by Matt Menez

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1) ENTERPRISE PROJECT PROPOSAL OVERVIEW

1.1) Full Topology Network Map

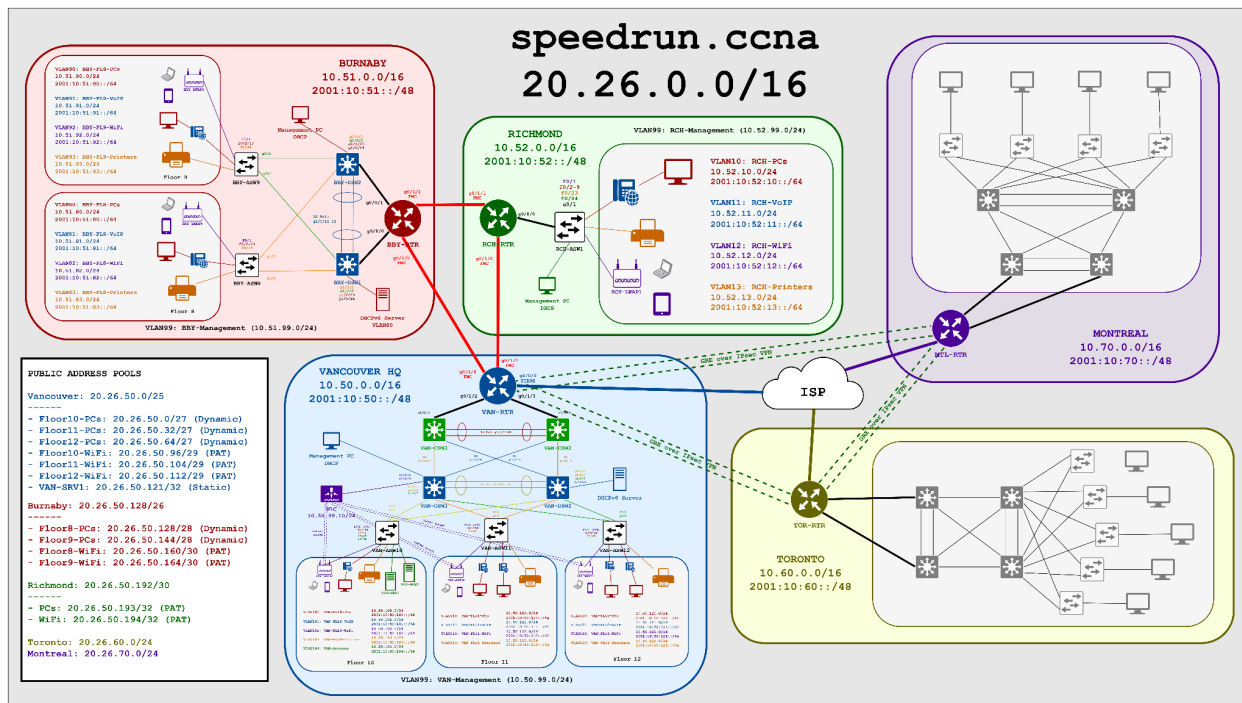


Figure 1.1: Full Topology Network Map for Speedrunning Inc.
(Don't worry. You'll go over each network in detail)

1.2) Speedrunning Inc. Strategic Project Overview

Speedrunning Incorporated is a growing IT services organisation in Canada, headquartered in Toronto, Ontario, with an already established office in Montreal, Quebec. With over 10000 annual clients and diverse departmental needs including help desk, networking, sysadmin, and cybersecurity, the organisation is looking to expand their services to the West Coast, which will begin in the Metro Vancouver region in British Columbia. As a result, Speedrunning Inc. requires an enterprise-level infrastructure that is secure, scalable, and highly available.

This proposal covers the network infrastructure according to Speedrunning's RFP requirements for a secure, scalable, and highly available regional offices in Metro Vancouver. The key objectives are:

- A complex and converged architecture spanning Vancouver, Burnaby, and Richmond
- Segmented, scalable LANs per device type and floor with consistent edge access control policies
- Seamless internal interconnectivity providing high-speed, dynamic routing between Vancouver, Burnaby, and Richmond
- A secure and private cross-country overlay interconnectivity to Toronto and Montreal
- Highly available and fault tolerant solutions deployed across core routing and switching network devices
- Centralised identity and access management paired with control-plane infrastructure hardening solutions
- Network services automating infrastructure synchronisation, dynamic host addressing, proactive auditing, and system logging
- A secure and hybrid (centralised and local) wireless tailored to the mobility and security needs of each office.

Each of these objectives are tested and documented in a proof of concept lab utilising Cisco Packet Tracer, GNS3, and EVE-ng.

Cisco network devices will be used for the proof of concept:

Router: Cisco ISR 4321

Multilayer Switch: 28-port Cisco Catalyst 3650

Layer 2 Access Switch: 48-port Cisco Catalyst 2960

1.3) Proposed Solutions

To meet Speedrunning Inc's requirements, the Speedrunning IT Department proposes a secure, scalable, fault tolerant, and converged network infrastructure. This solution combines foundational and advanced networking practises with a modern network security framework and seamless integration of different device types. This design prioritises security, scalability, and reliability, ensuring consistent user experience and operational efficiency across all Speedrunning Inc. locations.

1.3.a) Vancouver HQ LAN Overview and Requirements

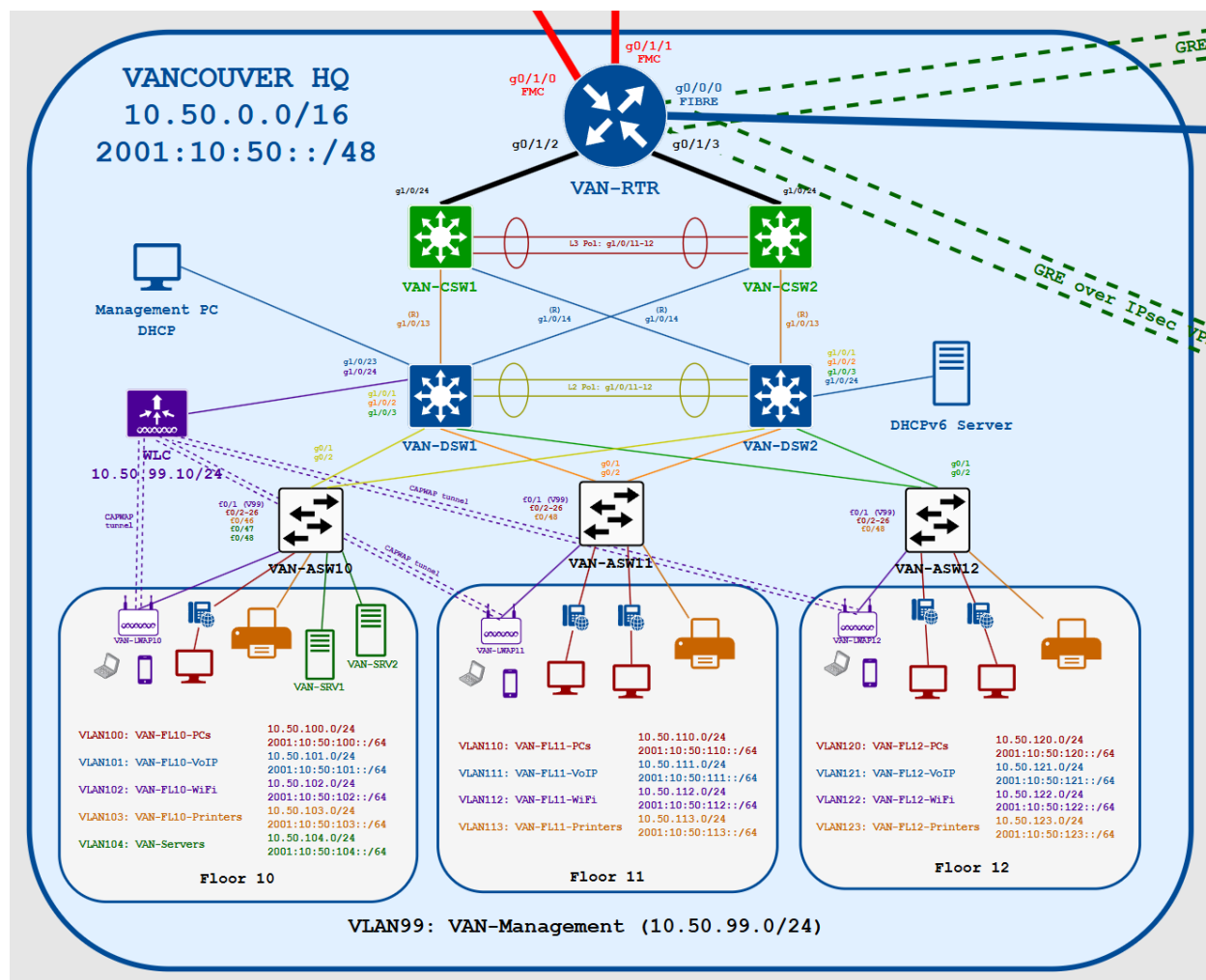


Figure 1.2: Vancouver HQ Office Full Topology

The Vancouver HQ office will serve as a co-headquarters for the West Coast alongside the Toronto HQ for the East Coast. It will be the largest office among the Metro Vancouver offices. Speedrunning Inc. will lease three consecutive floors (Floors 10, 11, 12) at Park Place at 666 Burrard Street in Downtown Vancouver.

The Vancouver HQ is expected to expand to six floors within the next three years. To address the scalability for a rigorous expansion deadline, this office will utilise a hierarchical three-tier network architecture with core, distribution, and access layers. This approach ensures modular scalability, high availability, and convenient troubleshooting domains required for enterprise operations.

Floor 10 will host the core routing and distribution networking hardware. Local and wide area traffic to and from the Vancouver office will terminate into the core edge router, VAN-RTR. Vancouver's router will also be responsible for providing seamless private connectivity to the remote networks in Toronto and Montreal. The Wireless LAN Controller, which provides the centralised Wi-Fi service for this office, will also be located on this floor.

Each floor will support 25 workstation clients, which will each be segmented into their own local network. Voice communication over IP phones will be supported and separated from standard data traffic. A high-capacity network printer will be deployed on each floor, with access being carefully controlled to prevent unauthorised network access. Centrally switched Wi-Fi will also be provided for wireless clients including work laptops, tablets, and personal BYOD devices.

To power the administrative services for the Metro Vancouver deployment, Floor 10 will host Speedrunning Inc's dedicated Linux server block. Operating as a centralised services hub, this infrastructure will extend its services to support the smaller networks in Burnaby and Richmond. The server block consists of the following:

- **VAN-SRV1:** Provides Metro Vancouver's web server, along with centralised DNS resolution and AAA RADIUS authentication for secure device management across all three offices.
- **VAN-SRV2:** Provides the SNMP manager and centralised system logging server to collect real-time alerts across the regional network.
- **DHCPv6 Server:** Connected to VAN-DSW2, this server provides local stateful DHCPv6 service to the Vancouver Office. DHCPv6 service will be decentralised, meaning Burnaby and Richmond will have their own local server suiting their addressing needs.

1.3.b) Burnaby LAN Overview and Requirements

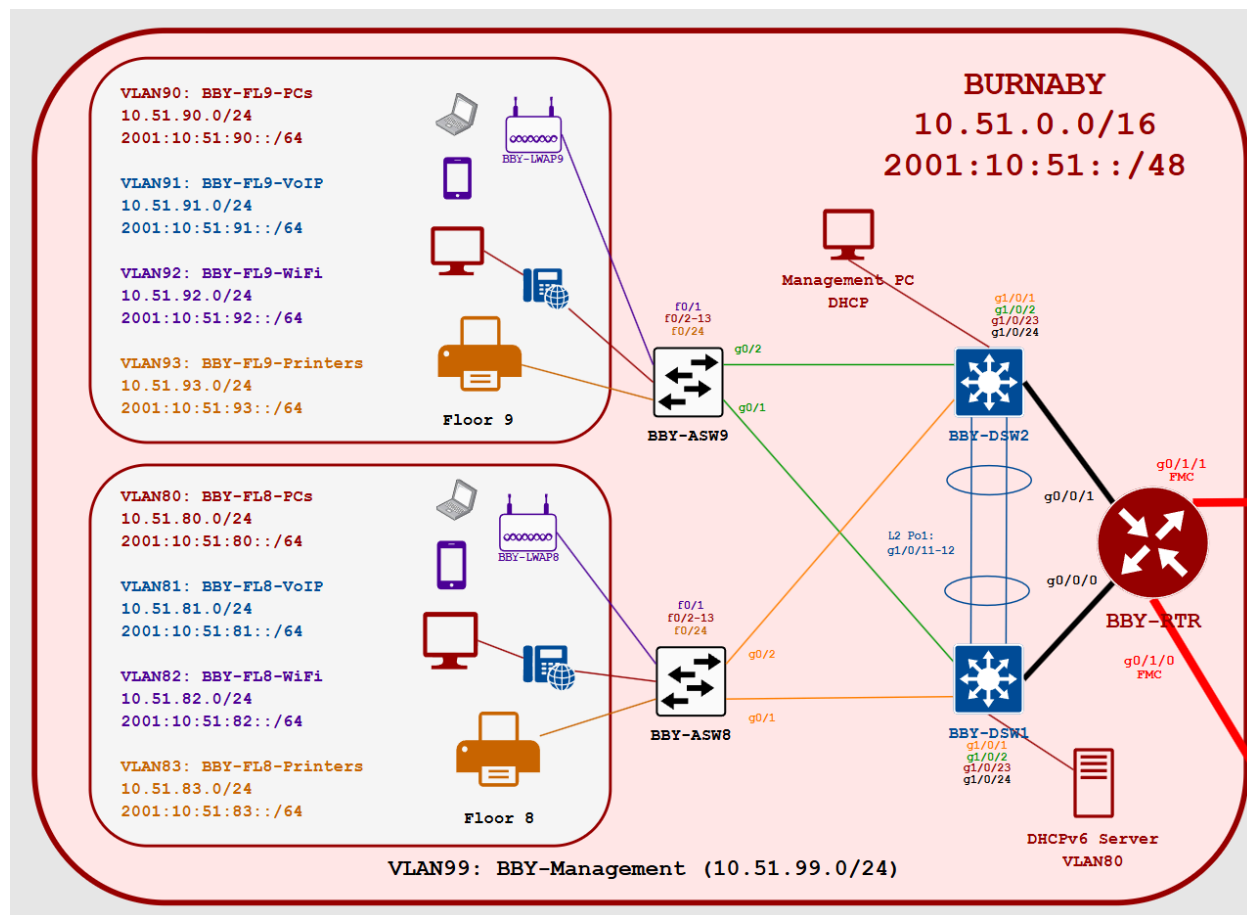


Figure 1.3: Burnaby Office Full Topology

The Burnaby branch office is one of the new branches along with Richmond. Speedrunning Inc. will lease two consecutive floors (Floors 8, 9) at Metrotower II at 4720 Kingsway in Metrotown, Burnaby.

Burnaby is expected to expand to three floors within the next year. To address the necessary scalability while addressing the office's medium size, this office will utilise a two-tier collapsed core network architecture with distribution and access layers. This approach ensures budget efficiency, modular scalability, high availability, and convenient troubleshooting domains required for enterprise long-term operations.

Floor 8 will host the collapsed core routing and distribution networking hardware. Local and wide area traffic to and from the Burnaby office will terminate into the core edge router, BBY-RTR. Wireless traffic will not be centrally switched to Vancouver's WLC, nor will a local WLC be deployed. Therefore, FlexConnect will be configured to allow the access points to locally switch their respective networks. This prevents bandwidth bottlenecks over the regional network.

Each floor will support 12 workstation clients, which will each be segmented into their own local network. Voice communication over IP phones will be supported and separated from standard data traffic. A high-capacity network printer will be deployed on each floor, with access being carefully controlled to prevent unauthorised network access. Locally switched Wi-Fi will also be provided for wireless clients including work laptops, tablets, and personal BYOD devices.

A DHCPv6 server will be connected to BBY-DSW1, providing local stateful DHCPv6 service to the Burnaby office.

1.3.c) Richmond LAN Overview and Requirements

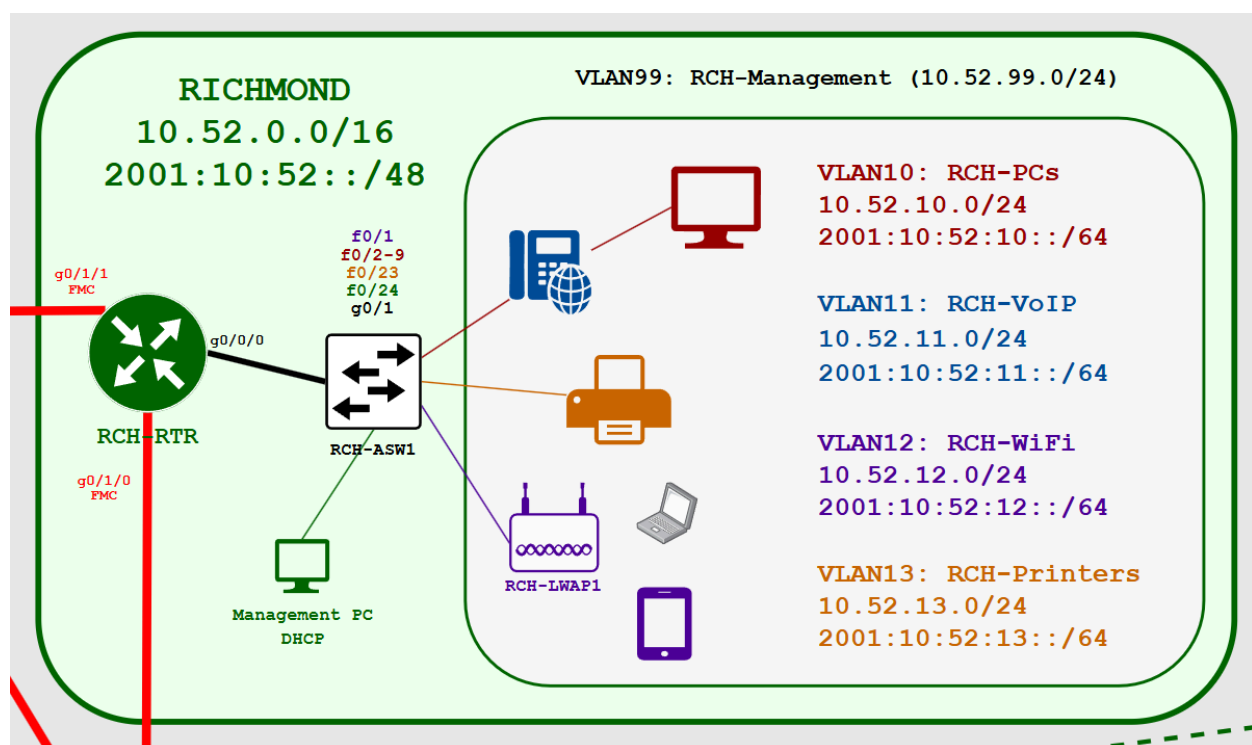


Figure 1.4: Richmond Office Full Topology

The Richmond branch office is one of the new branches along with Burnaby. Speedrunning Inc. will lease one office at the Cargo Village within the premises of YVR Airport.

Richmond is the smallest and most specialised office of the new three in Metro Vancouver. It will be catered to providing IT services within the airport. Due to the unique high-pressure nature of the airport's IT infrastructure, only a select team of twelve senior IT technicians of varying shifts will be stationed here. Therefore, a Small Office Home Office (SOHO) network plan will be deployed.

The SOHO office will host the core router, one switch, and the edge devices all at once. Local and wide area traffic to and from the Richmond office will terminate into the core edge router, RCH-RTR. Wireless traffic will not be centrally switched to Vancouver's WLC, nor will a local WLC be deployed. Therefore, FlexConnect will be configured to allow the access point to locally switch its respective network. This prevents bandwidth bottlenecks over the regional network.

The office will host eight workstations. Voice communication over IP phones will be supported and separated from standard data traffic. A high-capacity network printer will be deployed within the office, with access being carefully controlled to prevent unauthorised network access. Locally switched Wi-Fi will also be provided for wireless clients including work laptops, tablets, and personal BYOD devices.

The Richmond core router will be providing local stateful DHCPv6 service.

1.4) Remote LANs

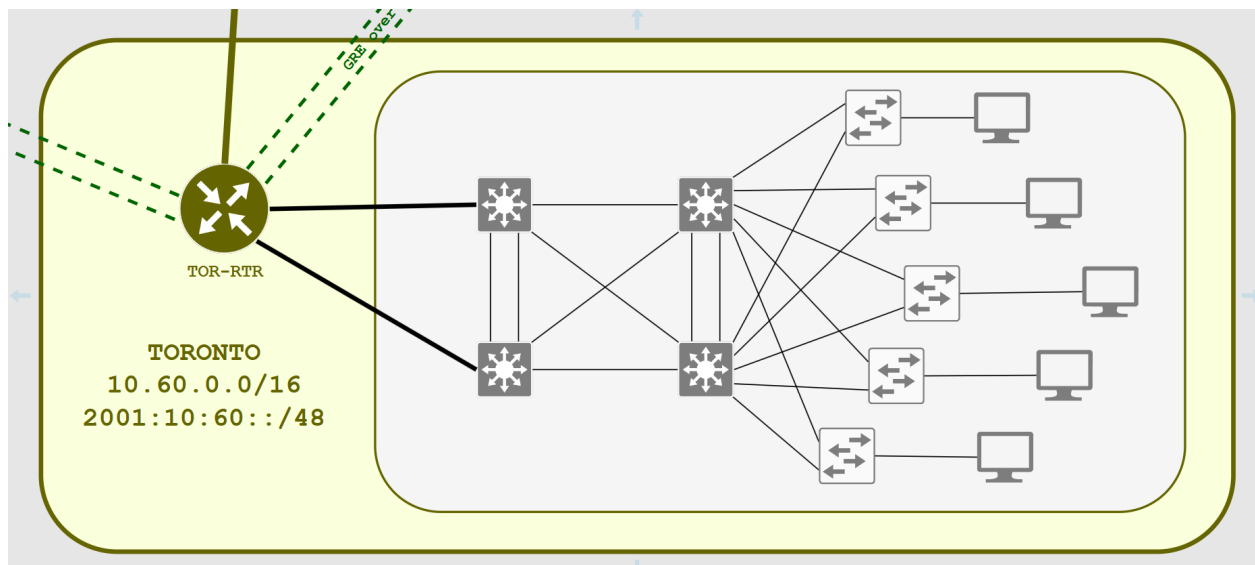


Figure 1.5: Toronto HQ Office Topology. LAN is greyed out for it is out of scope of this development.

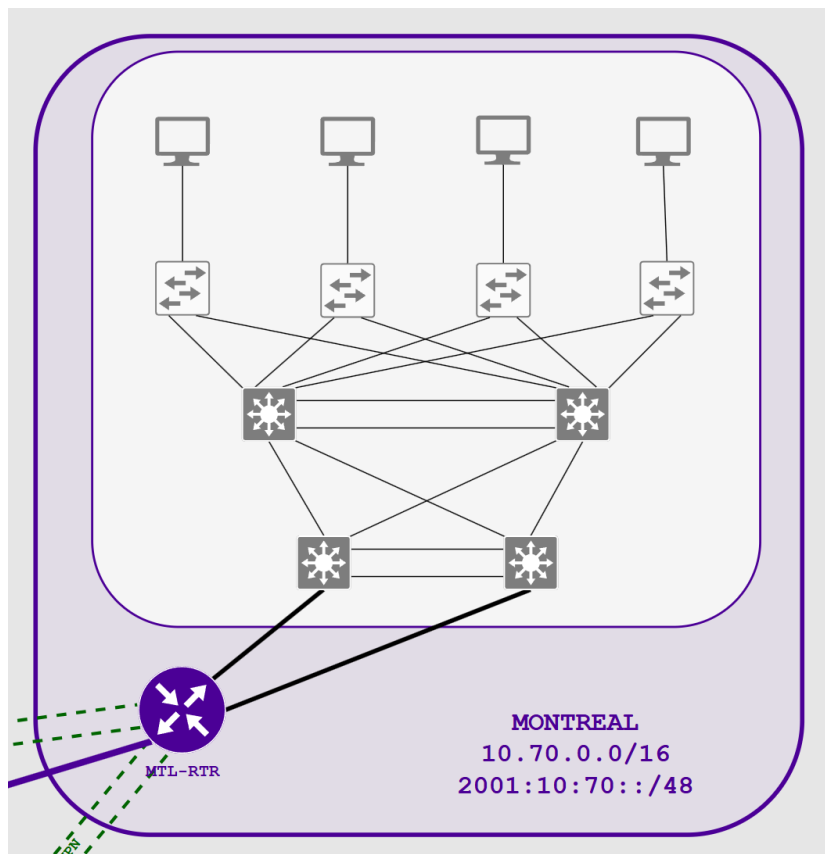


Figure 1.6: Montreal HQ Office Topology. LAN is greyed out for it is out of scope of this development.

Speedrunning Inc. has two established offices in Toronto and Montreal respectively. As requested by the organisation, the Metro Vancouver region must have a private and secure connection to these remote networks.

A **virtual private network (VPN)** will be implemented in the Vancouver HQ router to securely bridge an overlay connection to the remote Toronto and Montreal networks. Both the Toronto and Montreal routers are already pre-configured to establish this private tunnel. All that is left to do is for Vancouver to configure their side of the VPN to fully activate the secure connection.

2) SWITCHING ARCHITECTURE AND EDGE HARDENING

2.1) Virtual LAN Segmentation and Access Switch Port Assignments

Figure 2.1: VLAN segmentation for each office

X = Floor Number

Site	VLAN ID	VLAN Name	Description
VANCOUVER (Floors 10,11,12)	100, 110, 120	VAN-FLX-PCs	Workstations
	101, 111, 121	VAN-FLX-VoIP	IP Phones
	102, 112, 122	VAN-FLX-WiFi	Wi-Fi Clients
	103, 113, 123	VAN-FLX-Printers	Network Printers
	104	VAN-Servers	Regional Servers
	99	VAN-Management	Network Device Management
BURNABY (Floors 8,9)	80, 90	BBY-FLX-PCs	Workstations
	81, 91	BBY-FLX-VoIP	IP Phones
	82, 92	BBY-FLX-WiFi	Wi-Fi Clients
	83, 93	BBY-FLX-Printers	Network Printers
	99	BBY-Management	Network Device Management
RICHMOND	10	RCH-PCs	Workstations
	11	RCH-VoIP	IP Phones
	12	RCH-WiFi	Wi-Fi Clients
	13	RCH-Printers	Network Printers
	99	RCH-Management	Network Device Management

Figure 2.1: VLAN segmentation table for each office

Figure 2.2: Access port assignment for each access and distribution switch

Switch	Access interfaces	Assigned VLAN	Description
VAN-ASW10 VLANs: 99,100-104	f0/1	99	To LWAP, no FlexConnect
	f0/2-26	100 (data) 101 (voice)	To PCs with IP phones
	f0/46	103	To Printer
	f0/47	104	To VAN-SRV1
	f0/48	104	To VAN-SRV2
	f0/27-45	555	Administratively shut down unused interfaces
VAN-ASW11 VLANs: 99,110-113	f0/1	99	To LWAP, no FlexConnect
	f0/2-26	110 (data) 111 (voice)	To PCs with IP phones
	f0/48	113	To Printer
	f0/27-47	555	Administratively shut down unused interfaces
VAN-ASW12 VLANs: 99,120-123	f0/1	99	To LWAP, no FlexConnect
	f0/2-26	120 (data) 121 (voice)	To PCs with IP phones
	f0/48	123	To Printer
	f0/27-47	555	Administratively shut down unused interfaces
VAN-DSW1 VLANs: all used VLANs	g1/0/23	99	To Management PC
	g1/0/4-10, g1/0/15-22, g1/1/1-4	555	Administratively shut down unused interfaces
VAN-DSW2 VLANs: all used VLANs	g1/0/24	104	To DHCPv6 Server
	g1/0/4-10, g1/0/15-23, g1/1/1-4	555	Administratively shut down unused interfaces
BBY-ASW8 VLANs: 80-83,99	f0/2-13	80 (data) 81 (voice)	To PCs with IP phones
	f0/24	83	To Printer

	f0/14-23	555	Administratively shut down unused interfaces
BBY-ASW9 VLANs : 90-93,99	f0/2-13	90 (data) 91 (voice)	To PCs with IP phones
	f0/24	93	To Printer
	f0/14-23	555	Administratively shut down unused interfaces
VAN-DSW1 VLANs : all used VLANs	g1/0/23	80	To DHCPv6 Server
	g1/0/3-10, g1/0/13-22, g1/1/1-4	555	Administratively shut down unused interfaces
VAN-DSW2 VLANs : all used VLANs	g1/0/23	99	To Management PC
	g1/0/3-10, g1/0/13-22, g1/1/1-4	555	Administratively shut down unused interfaces
RCH-ASW1 VLANs : 10-13,99	f0/2-9	10 (data) 11 (voice)	To PCs with IP phones
	f0/23	13	To Printer
	f0/24	99	To Management PC
	f0/10-22, g0/2	555	Administratively shut down unused interfaces

Figure 2.2: Access port assignment for each access and distribution switch

Virtual LANs are implemented to contain broadcast domains and to provide network security and efficiency. Each device type is separated into a VLAN: PCs, IP phones, printers, servers, and wireless devices. Voice traffic is granted its own broadcast domain to facilitate Quality of Service (QoS) tagging. The Management VLAN is completely inaccessible to standard user endpoints.

The created VLANs are assigned certain ports on all access switches, as shown by the Figure 2.2 table above. All unused switch ports are parked in a "black hole" VLAN 555 and are **administratively shut down** to prevent unauthorised network access and malware tool manipulation.

Vancouver wireless Access Points (APs) are **centrally managed**, meaning data traffic is forwarded to a Wireless LAN Controller (WLC). Data traffic will be encapsulated within management traffic using CAPWAP. Therefore, an access port on VLAN 99 will be configured on the connecting switch ports.

Burnaby and Richmond APs will use **FlexConnect**, a wireless solution allowing branch access points to locally switch data traffic instead of sending it back to the WLC. Because of this feature, a trunk port allowing the respective Wi-Fi and Management VLANs will be configured on the connecting switch ports.

2.2) Inter-switch Connectivity (802.1Q VLAN Trunking)

Figure 2.3: Trunk port assignment for each access and distribution switch

Switch	Trunk interfaces	Assigned VLANs	Description
VAN-ASW10	g0/1-2	100-104,99,999 Native: 999	Trunk to Distribution Switches
VAN-ASW11	g0/1-2	110-113,99,999 Native: 999	Trunk to Distribution Switches
VAN-ASW12	g0/1-2	120-123,99,999 Native: 999	Trunk to Distribution Switches
VAN-DSW1 VAN-DSW2	Pol (g1/0/11-12)	100-104,110-113, 120-123,99,999 (ALL VLANs) Native: 999	Peer connection to partner distro switch
	g1/0/1	100-104,99,999 Native: 999	Trunk to VAN-ASW10
	g1/0/2	110-113,99,999 Native: 999	Trunk to VAN-ASW11
	g1/0/3	120-123,99,999 Native: 999	Trunk to VAN-ASW12
	g1/0/24 (VAN-DSW1)	99,102,112,122 Native: 99	To WLC
BBY-ASW8	f0/1	82,99 Native: 99	To LWAP with FlexConnect
	g0/1-2	80-83,99,999 Native: 999	Trunk to Distribution Switches
BBY-ASW9	f0/1	92,99 Native: 99	To LWAP with FlexConnect
	g0/1-2	90-93,99,999 Native: 999	Trunk to Distribution Switches
BBY-DSW1 BBY-DSW2	Pol (g1/0/11-12)	80-83,90-93,99, 999 (ALL VLANs) Native: 999	Peer connection to partner distro switch
	g1/0/1	80-83,99,999 Native: 999	Trunk to BBY-ASW8
	g1/0/2	90-93,99,999	Trunk to BBY-ASW9

		Native: 999	
RCH-ASW1	f0/1	12,99 Native: 99	To LWAP with FlexConnect
	g0/1	10-13,99,999 Native: 999	Trunk to Router

Figure 2.3: Trunk port assignment for each access and distribution switch

Uplinks from the access layer to the distribution layer and peer links between distribution switches are configured as **trunk ports**. Trunk ports can carry frame traffic for multiple VLANs across a single physical link. Any untagged frame traffic (traffic without a 802.1Q VLAN tag) will default to the native VLAN 999.

To accommodate all wireless networks at the Vancouver HQ, an 802.1Q trunk port allowing all Wi-Fi VLANs will be configured on the port connecting to the WLC. Because the WLC's management interface handles untagged traffic, the Management VLAN 99 will be configured as the native VLAN to ensure seamless management connectivity.

2.3) High Availability Distribution Links (Layer 2 EtherChannel)

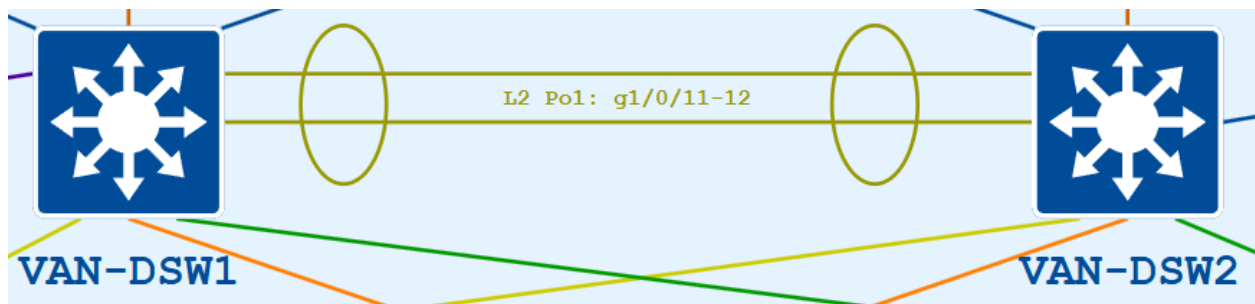
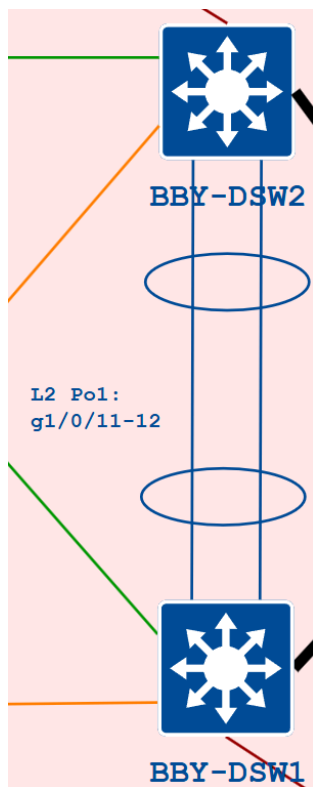


Figure 2.4: Layer 2 EtherChannel between peering Vancouver distribution switches



To achieve high availability, fault tolerance, and maximised bandwidth efficiency in the Layer 2 domain, **Layer 2 EtherChannel** will be deployed between peering distribution switches at the Vancouver HQ and Burnaby Branch offices.

These EtherChannels are trunk ports that can carry traffic within any VLAN in their respective office.

The Vancouver distribution switches will use **LACP**, an industry-standard protocol. The Burnaby distribution switches will use **PAgP**, a Cisco-proprietary EtherChannel protocol.

Figure 2.5: Layer 2 EtherChannel between Peering Burnaby distribution switches

2.4) Automated Topology Discovery (Link Layer Discovery Protocol)

Cisco network devices are automatically enabled with Cisco Discovery Protocol (CDP). Because CDP is a Cisco-proprietary standard, it will be disabled in preference for **Link Layer Discovery Protocol (IEEE 802.1AB)** across all network devices. This vendor-neutral approach future-proofs the network infrastructure, ensuring that Cisco devices can seamlessly discover and exchange information with non-Cisco network devices and endpoints.

2.5) Spanning Tree Protocol Optimization

2.5.a) Flavour Selection: Rapid PVST+

To prevent broadcast storms, the switching infrastructure will standardize on **Rapid Per-VLAN Spanning Tree Plus (Rapid PVST+)**. This choice ensures that all VLANs are running an independent STP instance, ensuring a loop-free network while allowing for rapid network convergence and per-VLAN load balancing.

2.5.b) Root Bridges for Different VLANs

Primary Root Bridges

VAN-DSW1: Floor 10 and Management VLANs

VAN-DSW2: Floor 11 and 12 VLANs

BBY-DSW1: Floor 8 and Management VLANs

BBY-DSW2: Floor 9 VLANs

Rapid PVST+ allows assignment of different root bridges for different VLANs. This load balancing strategy ensures that certain distribution switches are primary root bridges for specific VLANs. Any other VLANs not assigned as a primary root bridge and/or are the root bridge of the peer switch are explicitly configured as **secondary root bridges** on the local switch.

2.5.c) Edge Port Optimization (Portfast)

PortFast is configured on all ports directly connected to end devices to instantly transition the port to a **forwarding state**. This skips the listening and learning delays, ensuring that devices have immediate network access the moment they are plugged in.

2.5.d) STP Security Toolkit (BPDU Guard, Root Guard, Loop Guard)

By default, switches are trusting of any incoming **Bridge Protocol Data Units (BPDUs)**, which are messages exchanged between switches to determine the root bridge of the network. These messages are sent out even by brand new switches. To protect against malicious STP manipulation, accidental bridging loops, and rogue hardware deployments, STP security will be enforced at the access and distribution switches.

- **BPDU Guard:** Deployed on all PortFast-enabled access ports, interfaces configured with BPDU Guard will immediately shut down upon receipt of a BPDU. It will also place the interface into an administrative err-disable state. This protects the current STP topology from maliciously connected switches or endpoints pretending to be switches.
- **Root Guard:** Used on designated ports, interfaces configured with Root Guard will deny new superior BPDUs, preventing the switch from recognising new superior switches as the root bridge. This protects the root bridge hierarchy.
- **Loop Guard:** Switches rely on continuous receipt of BPDU messages to know about root bridges in real time. Used on non-designated ports, interfaces configured with Loop Guard will lock the interface into a loop-inconsistent blocking state upon no receipt of a BPDU after a given time. The blocking state will only be lifted if the physical bidirectional signal is restored. This protects the network from Layer 2 loops.

2.6) Edge Security Hardening

To secure the office networks in the Layer 2 level, a security toolkit combining port security, DHCP snooping, and dynamic ARP inspection will be deployed to secure endpoints from Layer 2 network attacks.

2.6.a) MAC-based Access Control (Switch Port Security)

Port security will be implemented on all access ports to maintain endpoint integrity and prevent MAC flooding attacks. All access ports will be allowed the **minimum** necessary number of MAC addresses to block any unauthorised devices from gaining network access. All invalid traffic from unauthorised MAC addresses will be discarded and logged to a syslog server. Switches will be configured with the **sticky** binding feature to automatically learn the MAC address of the first plugged device and save it to the running configuration.

PC links will limit up to 2 MAC addresses (1 for the PC and phone each); **FlexConnect AP links** will allow up to 20 wireless clients, and all other links to other devices will be limited to only 1 MAC address.

2.6.b) Rogue DHCP Mitigation (DHCP Snooping)

DHCP Snooping will be enabled for all active VLANs to protect them from rogue DHCP servers and malicious DHCP-based attacks such as DHCP starvation and DHCP poisoning. DHCP snooping works by only trusting ports connected to a trusted DHCP server and leaving the rest of the ports untrusted. DHCP snooping will then drop any unauthorised DHCP server traffic on untrusted ports and cross-references user traffic against a dynamic binding table to prevent IP and MAC spoofing. Rate limiting will also be enabled to prevent attackers from overwhelming the switch.

2.6.c) Man-in-the-middle Defence (Dynamic ARP Inspection)

Dynamic ARP Inspection will be enabled for all active VLANs to protect them from ARP-based attacks including ARP poisoning, better known as the **Man-in-the-middle** attack. ARP inspection works by only trusting ports connected to upstream network devices (such as the distribution switches) and DHCP servers. All end-user ports are left untrusted. ARP inspection will then inspect ARP traffic on the untrusted ports and validate them against the DHCP Snooping table. All mismatching ARP traffic will be discarded. Rate limiting will also be enabled to prevent attackers from overwhelming the switch.

3) ROUTING ARCHITECTURE, GATEWAY REDUNDANCY, TRAFFIC ACCESS CONTROL

3.1) High Availability Core Link (Layer 3 EtherChannel)

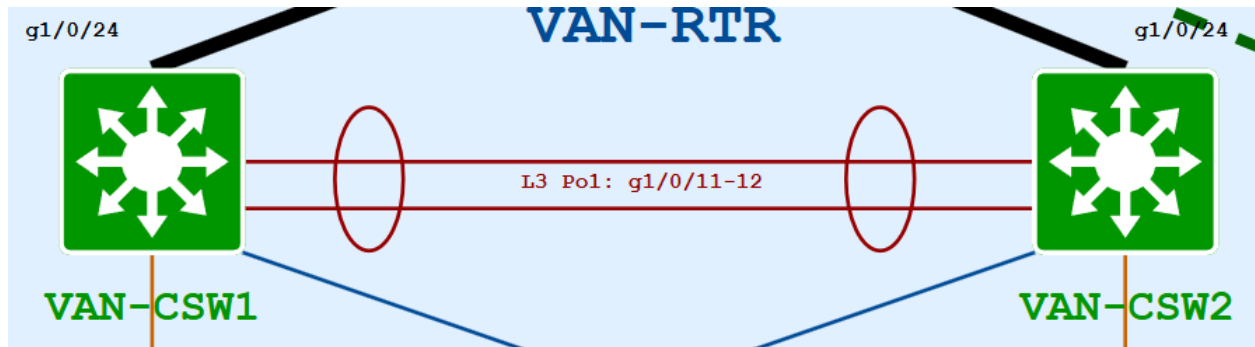


Figure 3.1: Layer 3 EtherChannel between peering Vancouver core switches

To achieve high availability, fault tolerance, and maximised bandwidth efficiency in the Layer 3 domain, **Layer 3 EtherChannel** will be deployed between peering core switches at the Vancouver HQ office. LACP will be used in the EtherChannel between the Vancouver core switches.

3.2) Dual-stack Addressing Architecture

3.2.a) Classless IPv4 Addressing Scheme

Figure 3.2: IPv4 Subnetting Table for each network

Site Location	Network description	Network address /subnet mask	Default Gateway
VANCOUVER HQ 10.50.0.0/16	Floor 10 Workstations	10.50.100.0/24	10.50.100.254
	Floor 10 IP Phones	10.50.101.0/24	10.50.101.254
	Floor 10 Wi-Fi Clients	10.50.102.0/24	10.50.102.254
	Floor 10 Printers	10.50.103.0/24	10.50.103.254
	Floor 10 Servers	10.50.104.0/24	10.50.104.254
	Floor 11 Workstations	10.50.110.0/24	10.50.110.254
	Floor 11 IP Phones	10.50.111.0/24	10.50.111.254
	Floor 11 Wi-Fi Clients	10.50.112.0/24	10.50.112.254
	Floor 11 Printers	10.50.113.0/24	10.50.113.254
	Floor 12 Workstations	10.50.120.0/24	10.50.120.254
	Floor 12 IP Phones	10.50.121.0/24	10.50.121.254
	Floor 12 Wi-Fi Clients	10.50.122.0/24	10.50.122.254
	Floor 12 Printers	10.50.123.0/24	10.50.123.254
	Management	10.50.99.0/24	10.50.99.254
BURNABY 10.51.0.0/16	Floor 8 Workstations	10.51.80.0/24	10.51.80.254
	Floor 8 IP Phones	10.51.81.0/24	10.51.81.254
	Floor 8 Wi-Fi Clients	10.51.82.0/24	10.51.82.254
	Floor 8 Printers	10.51.83.0/24	10.51.83.254
	Floor 9 Workstations	10.51.90.0/24	10.51.90.254
	Floor 9 IP Phones	10.51.91.0/24	10.51.91.254
	Floor 9 Wi-Fi Clients	10.51.92.0/24	10.51.92.254
	Floor 9 Printers	10.51.93.0/24	10.51.93.254
	Management	10.51.99.0/24	10.51.99.254

RICHMOND 10.52.0.0/16	Workstations	10.52.10.0/24	10.52.10.254
	IP Phones	10.52.11.0/24	10.52.11.254
	Wi-Fi Clients	10.52.12.0/24	10.52.12.254
	Printers	10.52.13.0/24	10.52.13.254
	Management	10.52.99.0/24	10.52.99.254

Figure 3.2: IPv4 Subnetting Table for each network

Each office is allocated a master /16 private IP block. This block is segmented into /24 subnets, ensuring that every VLAN is contained within its own broadcast domain.

The addressing scheme utilises a "**loose on the inside**" strategy, where unlike the conservative nature of public IP allocations, private addressing will be generously planned to maximise space and set foundations for future expansions.

Figure 3.3: IPv4 Interface Addressing Table for Vancouver devices

Device	Interface (description)	IP address/subnet mask
VAN-RTR	Loopback0	50.50.1.1/32
	Loopback1	50.50.10.10/32
	G0/0/0 (to ISP)	20.26.50.253/30
	G0/1/0 (to BBY-RTR)	192.168.0.6/30
	G0/1/1 (to RCH-RTR)	192.168.0.9/30
	G0/1/2 (to VAN-CSW1)	192.168.50.1/30
	G0/1/3 (to VAN-CSW2)	192.168.50.5/30
VAN-CSW1	Loopback0	50.50.2.1/32
	Loopback1	50.50.20.10/32
	Port-channel1 (to VAN-CSW2)	192.168.50.9/30
	G1/0/13 (to VAN-DSW1)	192.168.50.21/30
	G1/0/14 (to VAN-DSW2)	192.168.50.13/30
	G1/0/24 (to VAN-RTR)	192.168.50.2/30
VAN-CSW2	Loopback0	50.50.2.2/32
	Loopback1	50.50.20.20/32
	Port-channel1 (to VAN-CSW1)	192.168.50.10/30
	G1/0/13 (to VAN-DSW2)	192.168.50.25/30
	G1/0/14 (to VAN-DSW1)	192.168.50.17/30
	G1/0/24 (to VAN-RTR)	192.168.50.6/30
VAN-DSW1	Loopback0	50.50.3.1/32
	Loopback1	50.50.30.10/32
	G1/0/13 (to VAN-CSW1)	192.168.50.22/30
	G1/0/14 (to VAN-CSW2)	192.168.50.18/30
	SVI interfaces (All active VLANs)	10.50.x.252/24 (x=VLAN ID)
VAN-DSW2	Loopback0	50.50.3.2/32
	Loopback1	50.50.30.20/32

	G1/0/13 (to VAN-CSW2)	192.168.50.26/30
	G1/0/14 (to VAN-CSW1)	192.168.50.14/30
	SVI interfaces (All active VLANs)	10.50.x.253/24 (x=VLAN ID)
VAN-ASW10	VLAN 99 SVI	10.50.99.1/24
VAN-ASW11	VLAN 99 SVI	10.50.99.2/24
VAN-ASW12	VLAN 99 SVI	10.50.99.3/24
VAN-SRV1	NIC	10.50.104.1/24
VAN-SRV2	NIC	10.50.104.2/24
WLC	Management Interface	10.50.99.10/24
All end devices (except servers)	NIC	DHCP
HSRP Config	SVI interfaces (All active VLANs)	10.50.x.254/24 (Virtual IP)

Figure 3.3: IPv4 Interface Addressing Table for Vancouver devices

The 10.50.0.0/16 private CIDR range is assigned to the Vancouver HQ network for **internal endpoints**. This block provides more than enough room to segment /24 subnets for all standard user VLANs.

The 192.168.50.0/24 private CIDR block is **completely isolated** from user traffic and is reserved exclusively for **backbone routing infrastructure** between the core router, core switches, and distribution switches. It provides the point-to-point IP allocations needed for routing interfaces.

Figure 3.4: IPv4 Interface Addressing Table for Burnaby devices

Device	Interface (description)	IP address/subnet mask
BBY-RTR	Loopback0	51.51.1.1/32
	Loopback1	51.51.10.10/32
	G0/1/0 (to VAN-RTR)	192.168.0.5/30
	G0/1/1 (to RCH-RTR)	192.168.0.1/30
	G0/0/0 (to BBY-DSW1)	192.168.51.1/30
	G0/0/1 (to BBY-DSW2)	192.168.51.5/30
BBY-DSW1	Loopback0	51.51.3.1/32
	Loopback1	51.51.30.10/32
	G1/0/24 (to BBY-RTR)	192.168.51.2/30
	SVI interfaces (All active VLANs)	10.51.x.252/24 (x=VLAN ID)
BBY-DSW2	Loopback0	51.51.3.2/32
	Loopback1	51.51.30.20/32
	G1/0/24 (to BBY-RTR)	192.168.51.6/30
	SVI interfaces (All active VLANs)	10.51.x.253/24 (x=VLAN ID)
BBY-ASW8	VLAN 99 SVI	10.51.99.1/24
BBY-ASW9	VLAN 99 SVI	10.51.99.2/24
All end devices (except servers)	NIC	DHCP
HSRP Config	SVI interfaces (All active VLANs)	10.51.x.254/24 (Virtual IP)

Figure 3.4: IPv4 Interface Addressing Table for Burnaby devices

The 10.51.0.0/16 private CIDR range is assigned to the Burnaby network for **internal endpoints**. This block provides more than enough room to segment /24 subnets for all standard user VLANs.

The 192.168.51.0/24 private CIDR block is **completely isolated** from user traffic and is reserved exclusively for **backbone routing infrastructure** between the core router and distribution switches. It provides the point-to-point IP allocations needed for routing interfaces.

Figure 3.5: IPv4 Interface Addressing Table for Richmond devices

Device	Interface (description)	IP address/subnet mask
RCH-RTR	Loopback0	52.52.1.1/32
	Loopback1	52.52.10.10/32
	G0/0/0.10	10.52.10.254/24
	G0/0/0.11	10.52.11.254/24
	G0/0/0.12	10.52.12.254/24
	G0/0/0.13	10.52.13.254/24
	G0/0/0.99	10.52.99.254/24
	G0/1/0 (to VAN-RTR)	192.168.0.10/30
	G0/1/1 (to BBY-RTR)	192.168.0.6/30
RCH-ASW1	VLAN 99 SVI	10.52.99.1/24
All end devices (except servers)	NIC	DHCP

Figure 3.5: IPv4 Interface Addressing Table for Richmond devices

The 10.52.0.0/16 private CIDR range is assigned to the Richmond network for **internal endpoints**. This block provides more than enough room to segment /24 subnets for all standard user VLANs.

Given Richmond's smaller network size, the core router will utilise sub-interfacing through **Router on a Stick (ROAS)** to provide routing service to all VLANs within a single physical link.

3.2.b) IPv6 Global Unicast and Link-local Addressing Scheme

Figure 3.6: IPv6 Subnetting Table for each network

Site Location	Network description	Prefix/prefix mask
VANCOUVER HQ 10.50.0.0/16	Floor 10 Workstations	2001:10:50:100::/64
	Floor 10 IP Phones	2001:10:50:101::/64
	Floor 10 Wi-Fi Clients	2001:10:50:102::/64
	Floor 10 Printers	2001:10:50:103::/64
	Floor 10 Servers	2001:10:50:104::/64
	Floor 11 Workstations	2001:10:50:110::/64
	Floor 11 IP Phones	2001:10:50:111::/64
	Floor 11 Wi-Fi Clients	2001:10:50:112::/64
	Floor 11 Printers	2001:10:50:113::/64
	Floor 12 Workstations	2001:10:50:120::/64
	Floor 12 IP Phones	2001:10:50:121::/64
	Floor 12 Wi-Fi Clients	2001:10:50:122::/64
	Floor 12 Printers	2001:10:50:123::/64
BURNABY 10.51.0.0/16	Floor 8 Workstations	2001:10:51:80::/64
	Floor 8 IP Phones	2001:10:51:81::/64
	Floor 8 Wi-Fi Clients	2001:10:51:82::/64
	Floor 8 Printers	2001:10:51:83::/64
	Floor 9 Workstations	2001:10:51:90::/64
	Floor 9 IP Phones	2001:10:51:91::/64
	Floor 9 Wi-Fi Clients	2001:10:51:92::/64
	Floor 9 Printers	2001:10:51:93::/64
RICHMOND 10.52.0.0/16	Workstations	2001:10:52:10::/64
	IP Phones	2001:10:52:11::/64
	Wi-Fi Clients	2001:10:52:12::/64
	Printers	2001:10:52:13::/64

Figure 3.6: IPv6 Subnetting Table for each network

All networks will have an IPv6 addressing scheme to ensure a smooth transition between IPv4 and IPv6 infrastructure.

Each office is allocated a master /48 Global Unicast Address block. This block is segmented into /64 subnets, following IPv6 addressing standards and ensuring that every VLAN is contained within its own broadcast domain.

A lack of a Management VLAN in the IPv6 addressing scheme is intentional to prevent accidental global exposure and simplify troubleshooting scenarios. Management traffic will only be exclusive to IPv4 traffic.

Figure 3.7: IPv6 Interface Addressing Table for Vancouver devices

Device	Interface (description)	IP address/prefix mask
VAN-RTR	Loopback0	2001::1/128
	Link-local	fe80::1
	G0/0/0 (to ISP)	2001:db8:2026:50::6/64
	G0/1/0 (to BBY-RTR)	2001:192:168::5/127
	G0/1/1 (to RCH-RTR)	2001:192:168::6/127
	G0/1/2 (to VAN-CSW1)	2001:192:168:50::2/127
	G0/1/3 (to VAN-CSW2)	2001:192:168:50::4/127
VAN-CSW1	Loopback0	2001::50:1/128
	Link-local	fe80::50:1
	Port-channel1 (to VAN-CSW2)	2001:192:168:50::6/127
	G1/0/13 (to VAN-DSW1)	2001:192:168:50::c/127
	G1/0/14 (to VAN-DSW2)	2001:192:168:50::8/127
	G1/0/24 (to VAN-RTR)	2001:192:168:50::3/127
VAN-CSW2	Loopback0	2001::50:2/128
	Link-local	fe80::50:2
	Port-channel1 (to VAN-CSW1)	2001:192:168:50::7/127
	G1/0/13 (to VAN-DSW2)	2001:192:168:50::e/127
	G1/0/14 (to VAN-DSW1)	2001:192:168:50::a/127
	G1/0/24 (to VAN-RTR)	2001:192:168:50::5/127
VAN-DSW1	Loopback0	2001::50:3/128
	Link-local	fe80::50:3
	G1/0/13 (to VAN-CSW1)	2001:192:168:50::d/127
	G1/0/14 (to VAN-CSW2)	2001:192:168:50::b/127
	SVI interfaces (All active VLANs)	2001:10:50:x::ffffd/64 (x=VLAN ID)
VAN-DSW2	Loopback0	2001::50:4/128

	Link-local	fe80::50:4
	G1/0/13 (to VAN-CSW2)	2001:192:168:50::f/127
	G1/0/14 (to VAN-CSW1)	2001:192:168:50::9/127
	SVI interfaces (All active VLANs)	2001:10:50:x::ffff/64 (x=VLAN ID)
VAN-SRV1	NIC	2001:10:50:104::1/64
VAN-SRV2	NIC	2001:10:50:104::2/64
DHCPv6 Server	NIC	2001:10:50:104::aaaa/64
All end devices (except servers)	NIC	DHCP
HSRP Config	SVI interfaces (All active VLANs)	2001:10:50:x::ffff/24 (Virtual IP)

Figure 3.7: IPv6 Interface Addressing Table for Vancouver devices

The 2001:10:50::/48 Global Unicast Address network is assigned to the Vancouver HQ network for internal endpoints. This block provides more than enough room to segment /64 subnets for all standard user VLANs.

The 2001:192:168:50::/64 Global Unicast Address network is reserved exclusively for backbone routing infrastructure between the core router, core switches, and distribution switches. It provides the point-to-point IP allocations needed for routing interfaces.

The router and Layer 3 switches will be assigned a static link-local address to provide a fixed anchor point for network administration and IPv6 troubleshooting.

Figure 3.8: IPv6 Interface Addressing Table for Burnaby devices

Device	Interface (description)	IP address/subnet mask
BBY-RTR	Loopback0	2001::2/128
	Link-local	fe80::2
	G0/1/0 (to VAN-RTR)	2001:192:168::4/127
	G0/1/1 (to RCH-RTR)	2001:192:168::2/127
	G0/0/0 (to BBY-DSW1)	2001:192:168:51::2/127
	G0/0/1 (to BBY-DSW2)	2001:192:168:51::4/127
BBY-DSW1	Loopback0	2001::51:1/128
	Link-local	fe80::51:1
	G1/0/24 (to BBY-RTR)	2001:192:168:51::3/127
	SVI interfaces (All active VLANs)	2001:10:51:x::ffff/64 (x=VLAN ID)
BBY-DSW2	Loopback0	2001::51:2/128
	Link-local	fe80::51:2
	G1/0/24 (to BBY-RTR)	2001:192:168:51::5/127
	SVI interfaces (All active VLANs)	2001:10:51:x::fffe/64 (x=VLAN ID)
DHCPv6 Server	NIC	2001:10:51:80::aaaa/64
All end devices (except servers)	NIC	DHCP
HSRP Config	SVI interfaces (All active VLANs)	2001:10:51:x::ffff/64 (Virtual IP)

Figure 3.8: IPv6 Interface Addressing Table for Burnaby devices

The 2001:10:51::/48 Global Unicast Address network is assigned to the Burnaby network for internal endpoints. This block provides more than enough room to segment /64 subnets for all standard user VLANs.

The 2001:192:168:51::/64 Global Unicast Address network is reserved exclusively for backbone routing infrastructure between the core router, core switches, and distribution switches. It provides the point-to-point IP allocations needed for routing interfaces.

The router and Layer 3 switches will be assigned a static link-local address to provide a fixed anchor point for network administration and IPv6 troubleshooting.

Figure 3.9: IPv6 Interface Addressing Table for Richmond devices

Device	Interface (description)	IP address/subnet mask
RCH-RTR	Loopback0	2001::3/128
	Link-local	fe80::3
	G0/0/0.10	2001:10:52:10::ffff/64 Link-local: fe80::52:10
	G0/0/0.11	2001:10:52:11::ffff/64 Link-local: fe80::52:11
	G0/0/0.12	2001:10:52:12::ffff/64 Link-local: fe80::52:12
	G0/0/0.13	2001:10:52:13::ffff/64 Link-local: fe80::52:13
	G0/1/0 (to VAN-RTR)	2001:192:168::7/127
	G0/1/1 (to BBY-RTR)	2001:192:168::3/127
All end devices (except servers)	NIC	DHCP

Figure 3.9: IPv6 Interface Addressing Table for Richmond devices

The 2001:10:52::/48 Global Unicast Address network is assigned to the Richmond network for internal endpoints. This block provides more than enough room to segment /64 subnets for all standard user VLANs.

Given Richmond's smaller network size, the core router will utilise sub-interfacing through **Router on a Stick (ROAS)** to provide routing service to all VLANs within a single physical link.

The router interfaces will be assigned a static link-local address to provide a fixed anchor point for network administration and IPv6 troubleshooting.

3.3) First Hop Redundancy Protocols (HSRPv2 for IPv4 and IPv6)

```

VAN-DSW1(config-if)#do sh stand b
P indicates configured to preempt.
|
Interface    Grp  Pri  P State   Active      Standby      Virtual IP
-----
Vl99         99   110  P Active local      10.50.99.253 10.50.99.254
Vl100        100  110  P Active local      10.50.100.253 10.50.100.254
Vl100        1000 110  P Active local      FE80::50:4   FE80::5:73FF:FEA0:3E8
Vl101        101  110  P Active local      10.50.101.253 10.50.101.254
Vl101        1010 110  P Active local      FE80::50:4   FE80::5:73FF:FEA0:3F2
Vl102        102  110  P Active local      10.50.102.253 10.50.102.254
Vl102        1020 110  P Active local      FE80::50:4   FE80::5:73FF:FEA0:3FC
Vl103        103  110  P Active local      10.50.103.253 10.50.103.254
Vl103        1030 110  P Active local      FE80::50:4   FE80::5:73FF:FEA0:406
Vl104        104  110  P Active local      10.50.104.253 10.50.104.254
Vl104        1040 110  P Active local      FE80::50:4   FE80::5:73FF:FEA0:410
Vl110        110  100  Standby 10.50.110.253 local      10.50.110.254
Vl110        1100 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:44C
Vl111        111  100  Standby 10.50.111.253 local      10.50.111.254
Vl111        1110 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:456
Vl112        112  100  Standby 10.50.112.253 local      10.50.112.254
Vl112        1120 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:460
Vl113        113  100  Standby 10.50.113.253 local      10.50.113.254
Vl113        1130 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:46A
Vl120        120  100  Standby 10.50.120.253 local      10.50.120.254
Vl120        1200 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:4B0
Vl121        121  100  Standby 10.50.121.253 local      10.50.121.254
Vl121        1210 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:4BA
Vl122        122  100  Standby 10.50.122.253 local      10.50.122.254
Vl122        1220 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:4C4
Vl123        123  100  Standby 10.50.123.253 local      10.50.123.254
Vl123        1230 100  Standby FE80::50:4   local      FE80::5:73FF:FEA0:4CE
VAN-DSW1(config-if)#

```



Figure 3.10: Simulating HSRPv2 on IPv4 and IPv6 on a Cisco vIOS switch in EVE-ng.

The Cisco-proprietary Hot Standby Router Protocol version 2 (HSRPv2) will be used to provide first hop redundancy for all standard user VLANs in both IPv4 and IPv6. It lets routers share a single virtual IP and MAC address, ensuring that the default gateways are still active even if one of the physical switches fails.

HSRPv2 will be deployed on the distribution switches in the Vancouver HQ and Burnaby office. The **last usable address** will be used as the Virtual IP address for both IPv4 and IPv6.

The active router will be matched with that of their primary root bridge VLANs from STP, and the standby router with that of their secondary root bridge VLANs. This ensures that switching and routing traffic follow the same optimal path. Preemption will be enabled to allow routers with higher priority to automatically take over the active router role from a reboot.

Vancouver Virtual IPs: 10.50.x.254, 2001:10:50:x::ffff

Burnaby Virtual IPs: 10.51.x.254, 2001:10:51:x::ffff

3.4) Dynamic Interior Routing (OSPF)

A dynamic routing protocol like **Open Shortest Path First (OSPF)** eliminates the need for manual static routing configuration and easily adapts to any changes in the network. OSPF is widely used for it provides fast, automated, and scalable routing within private enterprise networks. In this case, OSPFv2 will be used for dynamic IPv4 routing, and OSPFv3 for dynamic IPv6 routing.

3.4.a) Multi-area OSPFv2

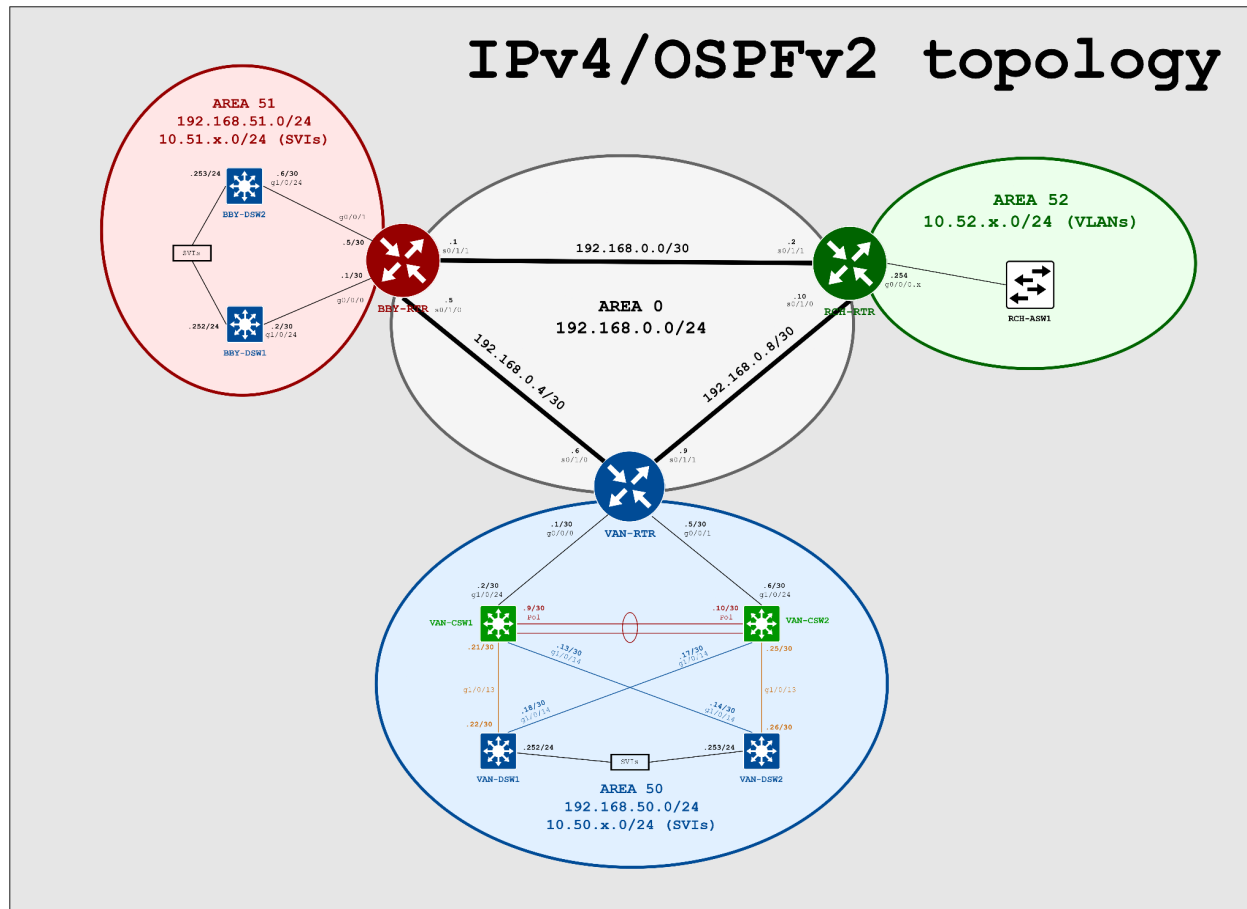


Figure 3.11: OSPFv2 Topology

Area 0: Core Router Backbone Area

Area 50: Vancouver LAN

Area 51: Burnaby LAN

Area 52: Richmond LAN

The OSPFv2 domain will be segmented into multiple areas to minimise routing table sizes and conserve router CPU and memory. A multi-area solution also localises the routing domains within the LAN boundaries, allowing for simpler administration and troubleshooting.

```

VAN-DSW1(config-if)#do sh ip route ospf
    10.0.0.0/8 is variably subnetted, 16 subnets, 2 masks
O IA    10.51.0.0 [110/6606] via 192.168.50.21, 00:00:42, GigabitEthernet1/0/13
        [110/6606] via 192.168.50.17, 00:00:42, GigabitEthernet1/0/14
O IA    10.52.0.0 [110/6596] via 192.168.50.21, 00:00:57, GigabitEthernet1/0/13
        [110/6596] via 192.168.50.17, 00:00:57, GigabitEthernet1/0/14
    50.0.0.0/32 is subnetted, 10 subnets

```

Figure 3.12: From Vancouver, OSPF routes to Burnaby (10.51.0.0) and Richmond (10.52.0.0) are summarised into the master /16 network instead of listing every active /24 network.

Summarise route commands

```

VAN-RTR(config-router)# area 50 range 10.50.0.0 255.255.0.0
BBY-RTR(config-router)# area 51 range 10.51.0.0 255.255.0.0
RCH-RTR(config-router)# area 52 range 10.52.0.0 255.255.0.0

```

Figure 3.13: The commands used to configure route summarisation on an Area Border Router.

Area 0 will provide the backbone area for the OSPF domain. All other areas will be directly connected to this area, and all routing information between different OSPF areas are exchanged here.

Areas 50, 51, and 52 each represent the three LANs connected to Area 0. Routing tables within the area are **localised**, meaning all local routes will be listed in full detail (no summaries). Conversely, routes to different areas will be **summarised** to simplify the routing table and allow the remote LAN to process the receiving traffic locally instead of the sender LAN.

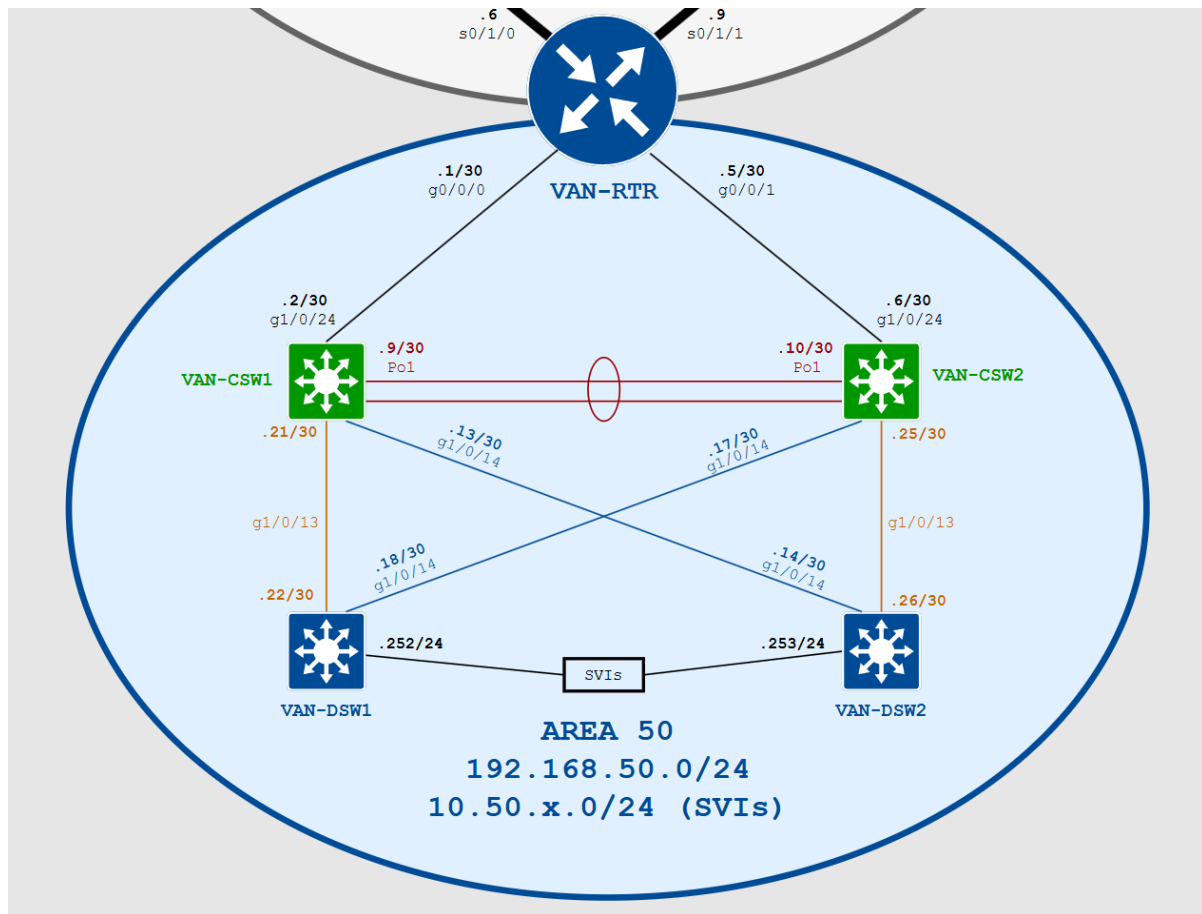


Figure 3.14: Close-up of Vancouver's routing map. All links between network devices are in a /30 network. In point-to-point networks, there are no Designated Routers or Backup Designated Routers.

All neighbouring links will use a point-to-point network type to eliminate Designated Router (DR) elections, reducing resource overhead and speeding up link convergence.

Excluding Management VLANs, OSPF will be passive on Loopback and SVI interfaces to reduce resource overhead. The cost reference bandwidth will be set to **10 Gbps** to differentiate between high-speed links.

3.4.b) Multi-area OSPFv3

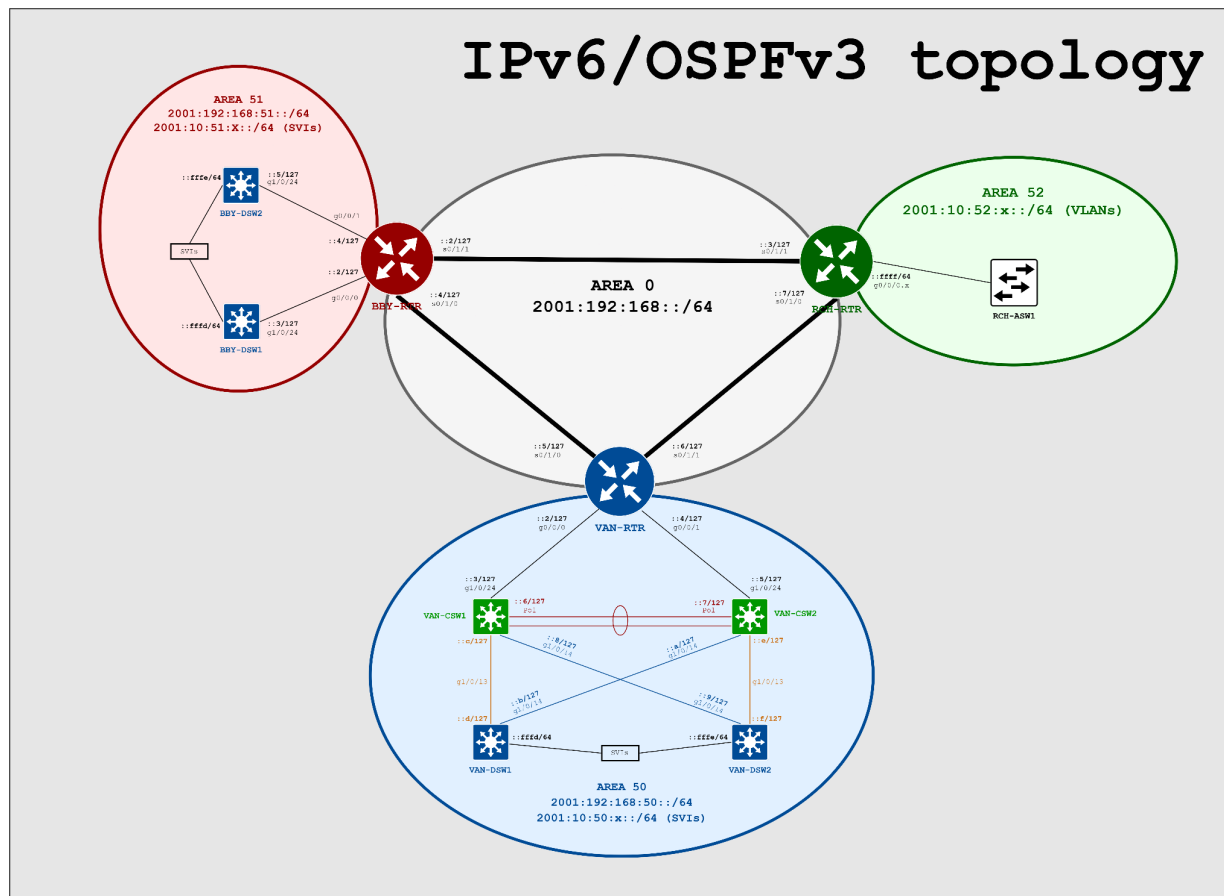


Figure 3.15: OSPFv3 Topology

Area 0: Core Router Backbone Area

Area 50: Vancouver LAN

Area 51: Burnaby LAN

Area 52: Richmond LAN

The OSPFv3 domain will be segmented into multiple areas to minimise routing table sizes and conserve router CPU and memory. A multi-area solution also localises the routing domains within the LAN boundaries, allowing for simpler administration and troubleshooting.

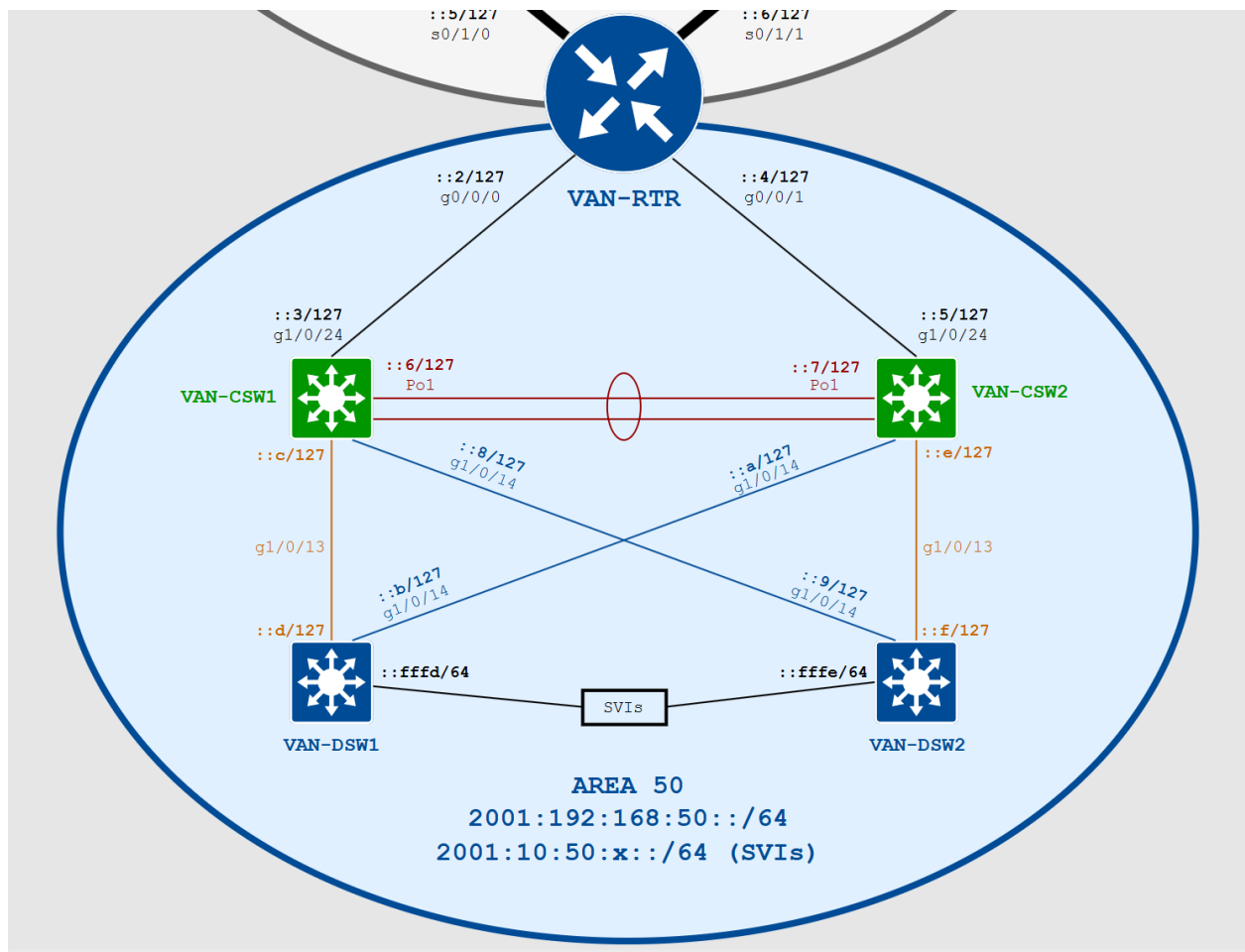


Figure 3.16: Close-up of Vancouver's routing map. Notice that all links between network devices are in a /127 network. In point-to-point networks, there are no Designated Routers or Backup Designated Routers.

All neighbouring links will use a point-to-point network type to eliminate Designated Router (DR) elections, reducing resource overhead and speeding up link convergence.

OSPF will be passive on Loopback and SVI interfaces to reduce resource overhead. The cost reference bandwidth will be set to **10 Gbps** to differentiate between high-speed links.

3.4.c) OSPF Authentication

```
VAN-RTR(config)#do sh ip ospf int g0/0/0

GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.50.1/30, Area 50
  Process ID 2, Router ID 50.50.1.1, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Index 5/5, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 50.50.2.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
VAN-RTR(config)#
```

Figure 3.17: MD5 will be used for authenticating OSPF neighbours.

OSPF authentication verifies the identity of neighbouring routers and ensures data integrity, allowing the exchange of routes in a secure and authenticated manner.

MD5 hashing will be used to authenticate neighbour-connected interfaces.

3.5) Static Routing

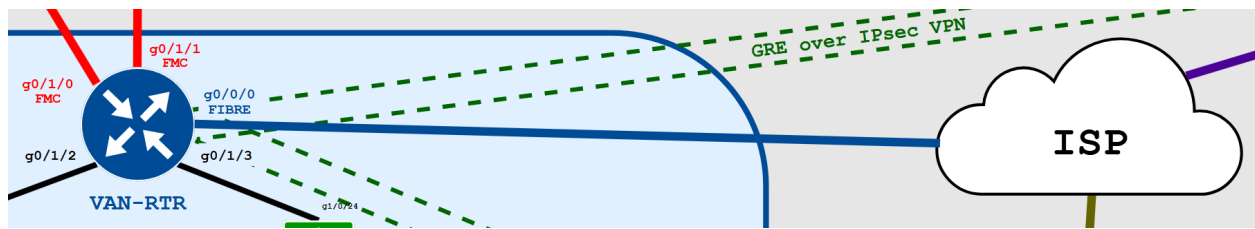


Figure 3.18: VAN-RTR's fibre connection to the ISP

An IPv4 and IPv6 default static route will be configured in VAN-RTR to provide internet access to the entire Metro Vancouver region. The default route will be advertised in the OSPF domain to tell all other routers and Layer 3 switches where to send traffic that doesn't match any specific route.

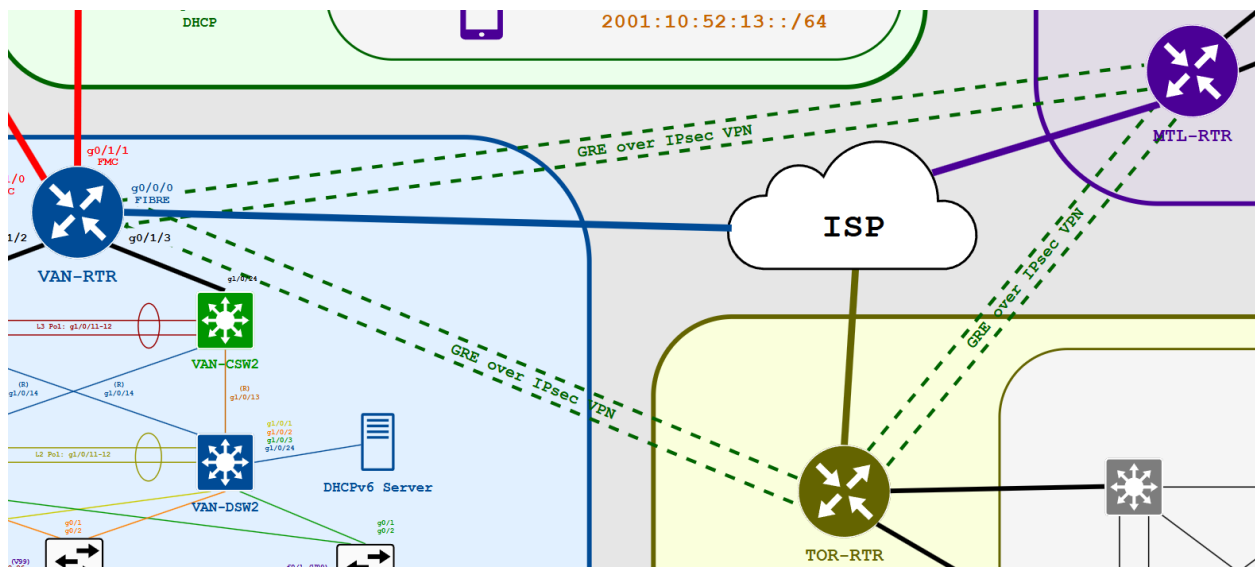


Figure 3.19: VAN-RTR's VPN connections to Toronto and Montreal

Two static routes, each one bound for **Toronto** (10.60.0.0/16) and **Montreal** (10.70.0.0/16), will be configured in VAN-RTR to bind outbound traffic to these remote offices through a system of secure VPN tunnels.

3.6) Edge Traffic Access Control Policies

To secure the office networks in the Layer 3 level, extended access control lists will be deployed to permit only the bare minimum network traffic required for devices to perform their tasks. All configured extended access lists will be applied **inbound** and **closest to the source** to prevent unwanted packets from traversing and wasting bandwidth across the network.

3.6.a) VoIP Phone Access Policy

To harden the voice infrastructure and protect the Cisco Unified Communication Manager Express (CUCME) ecosystem, IP phones are only permitted the following network traffic:

- **DHCP requests:** IP phones need to transmit DHCP requests to automatically obtain IP addresses.
- **All VoIP VLANs:** Network traffic will be unrestricted across all VoIP VLANs to ensure seamless voice communication between IP phones.
- **HSRPv2 protocol:** HSRPv2 uses the multicast address 224.0.0.102 and UDP port 1985 to transmit hello packets and other HSRP messages.
- **CUCME server:** Traffic to and from the CUCME server (VAN-RTR) is authorised to facilitate call processing and configurations.
- **Skinny Client Control Protocol (SCCP):** A lightweight Cisco-proprietary signaling protocol used to register IP phones.
- **Session Initiation protocol (SIP):** An open-standard signaling protocol used alongside SCCP to initiate, maintain, modify, and terminate voice sessions.
- **Real-time Transport Protocol (RTP) streams:** This protocol delivers audio and video over IP networks.

All other unmatched traffic will be denied due to the implicit deny rule automatically applied in the bottom of the access list.

3.6.b) Network Printer Access Policy

Network printers are highly vulnerable because they are notoriously difficult to keep fully patched. This access list will be centred around blocking printers from initiating most network traffic. To protect networks from infected printers, the printer VLANs are only permitted the following network traffic:

- **DHCP requests:** Printers need to transmit DHCP requests to automatically obtain IP addresses.
- **DNS:** Printers will be authorised to resolve hostnames of network computers and servers.

- **Syslog:** Printer events and alerts will be captured in the security audit logs.
- **ICMP reply:** Permitting ICMP echo replies allows for pinging printers to verify reachability while blocking printers from initiating ping sweeps to other subnets.
- **HSRPv2 protocol:** HSRPv2 uses the multicast address 224.0.0.102 and UDP port 1985 to transmit hello packets and other HSRP messages.
- **All established TCP protocols:** This rule allows for incoming traffic from **established** sessions. This allows user subnets to initiate print jobs over standard printing protocols (such as LPD on TCP port 515) while blocking printers from initiating new TCP connections.

All other unmatched traffic will be denied due to the implicit deny rule automatically applied in the bottom of the access list.

3.6.c) Workstation and Wireless Client Access Policy

This access list includes a mix of permitting and denying specific network traffic to and from workstations and Wi-Fi clients. It will permit standard services and Internet use while denying access to critical infrastructure.

Permit:

- **DHCP requests:** for automatically obtaining IP addresses
- **DNS:** to resolve hostnames on the Internet
- **HSRPv2 protocol:** allow HSRPv2 traffic to maintain gateway redundancy

Deny:

- **Management and VoIP VLANs:** Management and VoIP VLANs are completely isolated from regular user data traffic, protecting voice traffic and preventing brute-force login attacks against management interfaces.
- **192.168.0.0/16 range:** This range is used exclusively by backbone routing interfaces. Standard users do not need to interact with the routing fabric.
- **All router and L3 switch Loopback addresses:** Regular users do not need to access or test connectivity to routers.

An **implicit permit** rule will be applied to allow browsing of the Internet.

3.6.d) Management Access Policy

This standard access policy only permits **local management VLANs** to all network hardware. It is applied **inbound** on all virtual lines. Remote management VLANs are not permitted.

4) SECURING MANAGEMENT INFRASTRUCTURE AND REMOTE CONNECTIVITY

4.1) Centralised Identity and Access Management and Infrastructure Hardening

To protect network infrastructure from unauthorised and malicious access, a rigorous security framework is implemented across all network devices. This architecture combines local platform hardening and centralised IAM solutions using industry-standard authentication protocols, shifting away from device-specific management toward a fully audited control plane.

4.1.a) Authentication Failure Lockouts and Type 9 Hashing

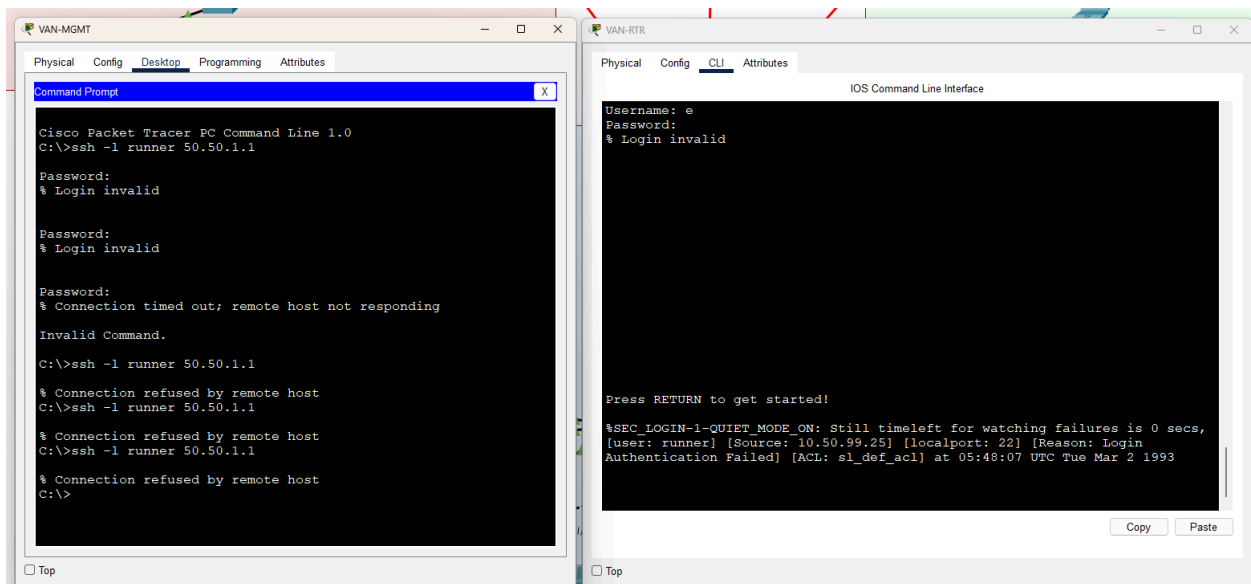


Figure 4.1: Authentication Failure Lockout in action on Packet Tracer. After three failed attempts within one minute, the Vancouver core router had denied the Management PC from attempting to login again for two minutes.

Network devices are configured to actively monitor failed login attempts. If a threshold is breached—defined as three failed attempts within 60 seconds—the targeted management line (VTY or Console) is completely blocked from login attempts for two minutes.

```

VAN-CSW1#sh run
Building configuration...

Current configuration : 5875 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname VAN-CSW1
!
login block-for 120 attempts 3 within 60
!
enable secret 9 $9$Jl9FIAftPZf7c3$HHZWqb9jOKKf0t1MNX0ls.DtyaVQSeVGuU.wLBq8L2k
!
!
!
!
!
no ip cef
ip routing
!
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
username runner secret 9 $9$Jl9FIAftPZf7c3$BhQumDGEBrLMacrmu.gIgtv6Kksc.D5QbHONR2cf8gI
!
!

```

Figure 4.2: Type 9 (scrypt) hashing on all passwords

Compatible network devices will encrypt all passwords with **Type 9 scrypt hashing**, which is designed to be highly resistant to modern GPU-based brute-force attacks.

Older switch and router models running legacy IOS versions are limited to **Type 5 MD5 hashing**, making them highly vulnerable to password attacks. However, authentication failure lockout still provides brute-force protection and real-time alerting, allowing IT personnel to isolate the offensive source IP before a breach can occur.

4.1.b) Centralised AAA RADIUS Authentication for Console Access

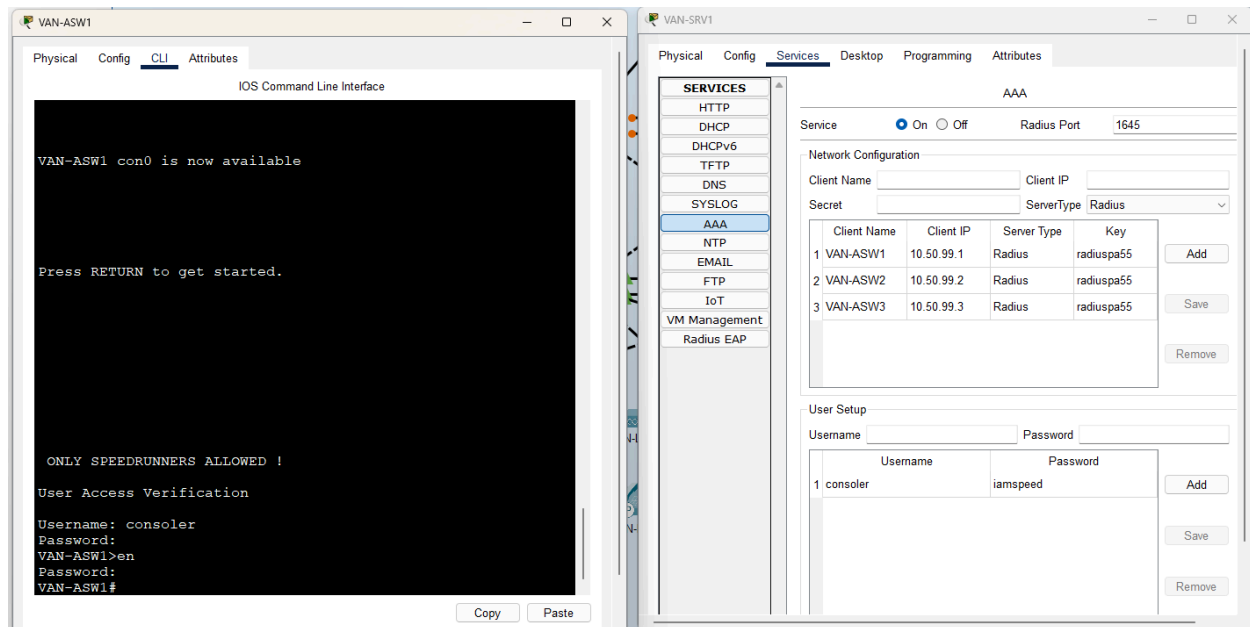


Figure 4.3: Logging in to a AAA-defined user. VAN-ASW1 is an authorised client of the RADIUS server.

```
# RADIUS CONFIGURATION (LEGACY)
VAN-ASW1(config)# aaa new-model
VAN-ASW1(config)# radius-server host 10.50.104.1
VAN-ASW1(config)# radius-server key radiuspa55
VAN-ASW1(config)# aaa authentication login CONSOLE-LOGIN group radius local

VAN-ASW1(config)# line con 0
VAN-ASW1(config-line)# login authentication CONSOLE-LOGIN
VAN-ASW1(config-line)#exec-timeout 30 0
VAN-ASW1(config-line)#logging sync

VAN-ASW1(config-line)#exit
VAN-ASW1(config)# wr
```

Figure 4.4: RADIUS authentication configuration on the console port. The **radius-server** commands are now deprecated and are replaced by the modern **radius server [server-name]** command format.

Relying on local passwords across a large network makes auditing impossible and increases breach risks. To solve this, the infrastructure enforces a **centralised AAA authentication** via the **RADIUS protocol** for all console connections. If the RADIUS server is offline, the authentication engine falls back to verifying the user against a local database on the device.

The console ports are configured to terminate the current session after **30 minutes of inactivity**, forcing the administrator to re-authenticate.

4.1.c) Local Authentication and Authorization for SSH Access

Secure Shell version 2 (SSHv2) will be used to securely manage local network devices and support stronger encryption algorithms. Cryptographic SSH keys will use the largest possible modulus size (**4096**).

All VTY lines will only allow **SSHv2**, and credentials are validated against the local configuration database. VTY sessions are terminated after **30 minutes of inactivity**, forcing the administrator to re-authenticate.

4.2) Secure Remote Connectivity (GRE over IPsec VPN)

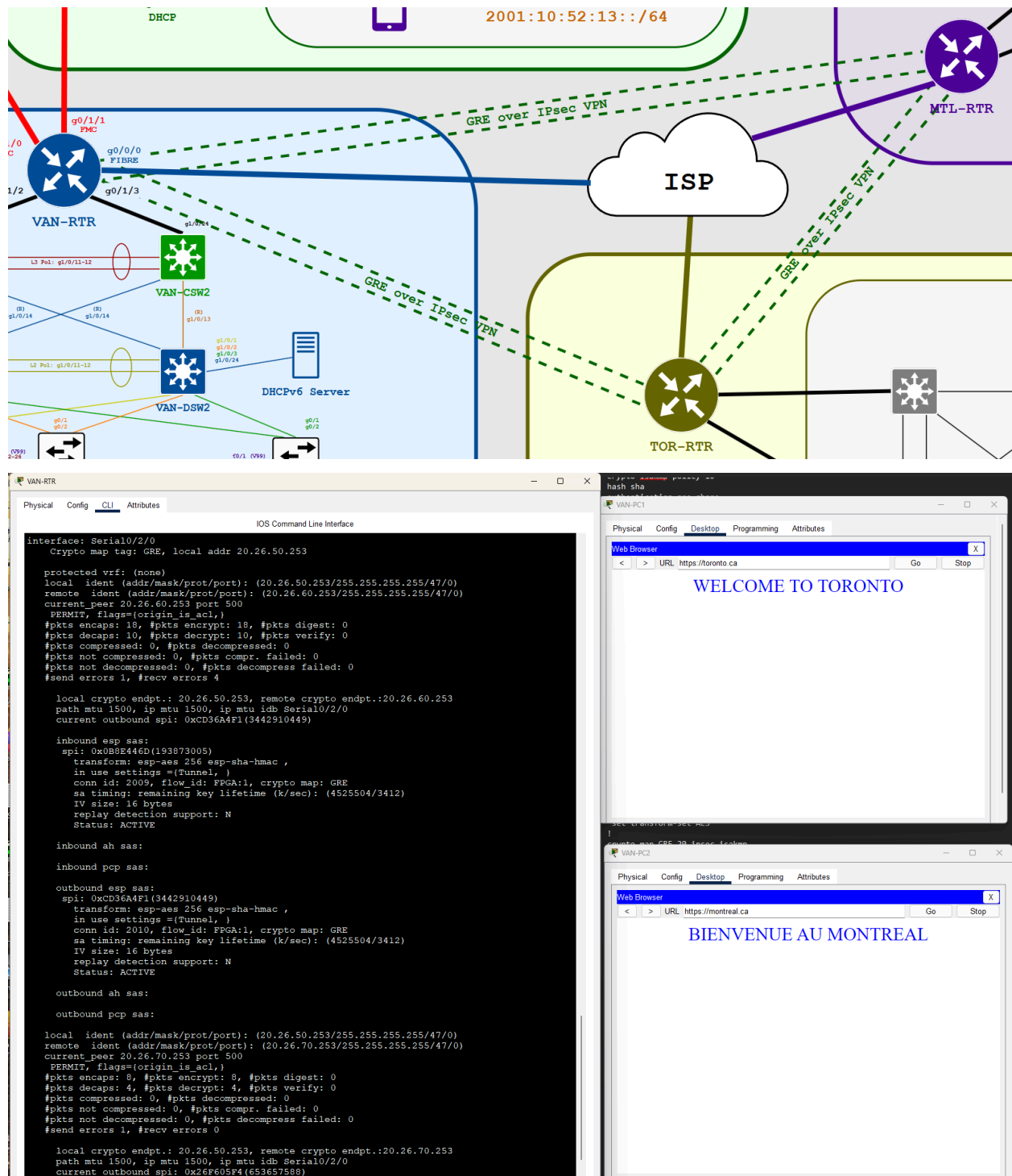


Figure 4.5: VAN-RTR's VPN connections to Toronto and Montreal. The second screenshot shows two Vancouver workstations gaining access to Toronto and Montreal's web servers through a VPN.

Vancouver needs a secure and reliable private connection to both Toronto and Montreal. To meet this requirement, a **virtual private network (VPN) infrastructure** will be deployed across these campuses.

The VPN itself will combine the use of **Generic Routing Encapsulation (GRE)** and **Internet Protocol Security (IPsec)** to encapsulate and encrypt outbound remote traffic across the Internet.

IPsec VPNs alone only support unicast traffic and are inflexible. To support multicast and broadcast traffic, as well as dynamic routing protocols, a **virtual GRE tunnel interface** will be established between Vancouver's router and its remote counterparts.

IPsec will then be used alongside GRE to secure these tunnels by providing robust encryption across the Internet. IPsec will apply advanced encryption such as AES-256 and strong hashing from SHA-256.

4.2.a) IKE Phase 1: Authenticating and Securing VPN Peers

```
VAN-RTR(config)#do sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
20.26.60.253 20.26.50.253 QM_IDLE        1021    0 ACTIVE
20.26.70.253 20.26.50.253 QM_IDLE        1009    0 ACTIVE

IPv6 Crypto ISAKMP SA

VAN-RTR(config)#
```

Figure 4.6: IKE Phase 1 verification on Cisco Router.

Before encrypting any data, the peer routers (Vancouver and Toronto/Montreal) must establish a secure channel to talk to one another. This is known as **ISAKMP (Phase 1)**.

The peers negotiate and agree on a strict security policy using the following parameters, known as **HAGLE**:

- **Hash:** SHA-256 is the most secure hashing function available.
- **Authentication:** Pre-shared keys are used to authenticate VPN peers.
- **Group (Diffie-Hellman):** Group 20 uses a 384-bit **Elliptic Curve Cryptography (ECC)**, which is extremely secure and faster in performance.
- **Lifetime:** Set to 86400 seconds (24 hours) before the keys are dynamically renegotiated.
- **Encryption:** AES-256 is the most secure encryption algorithm available.

4.2.b) IKE Phase 2: Securing the Data Transfer Channel

```
VAN-RTR(config)#
VAN-RTR(config)#do sh crypto ipsec sa

interface: Serial0/2/0
  Crypto map tag: GRE, local addr 20.26.50.253

  protected vrf: (none)
  local ident (addr/mask/prot/port): (20.26.50.253/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (20.26.60.253/255.255.255.255/47/0)
  current_peer 20.26.60.253 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 9

    local crypto endpt.: 20.26.50.253, remote crypto endpt.:20.26.60.253
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/2/0
    current outbound spi: 0x1A142723(437528355)

  inbound esp sas:
    spi: 0xAECE623B(2932761147)
--More--
```

Figure 4.7: IKE Phase 2 Verification on Cisco Router.

Once ISAKMP establishes a secure channel, the IPsec Security Association (Phase 2) begins. This phase builds the data tunnels that encrypt and transmit GRE-encapsulated corporate traffic.

Phase 2 follows a four-step sequence to define, filter, map, and apply the encryption policy:

- 1) **Defining the Transform Set:** The transform set defines the security protocols and cryptographic algorithms used to protect data flowing through the IPsec VPN tunnel. In this case, ESP will be used to encapsulate data, AES-256 for the encryption, and SHA with HMAC for hashing.
- 2) **Filtering GRE Traffic using ACLs:** Crypto ACLs define the exact protocol and source and destination IP addresses to trigger the VPN tunnel. This ACL permits GRE traffic between VAN-RTR's outside interface and the remote branch routers.
- 3) **Assembling the Crypto Map:** The crypto map is a set of rules that binds the transform set, VPN peers, and crypto ACLs together.
- 4) **Applying the Crypto Map:** Applying a crypto map to the outside interface tells the router what to encrypt (ACL), how to encrypt (transform set), and where to send the traffic to (VPN peers).

4.2.c) GRE Traffic Tunneling and Multi-Protocol Encapsulation

CONFIGURING GRE TUNNELS

```
VAN-RTR(config)# int tunnel0
VAN-RTR(config-if)# desc # to TOR-RTR #
VAN-RTR(config-if)# ip add 1.1.1.1 255.255.255.252
VAN-RTR(config-if)# tunnel source g0/0/0
VAN-RTR(config-if)# tunnel destination 20.26.60.253
VAN-RTR(config-if)# tunnel mode gre ip
VAN-RTR(config-if)# no shut

VAN-RTR(config-if)# int tunnel1
VAN-RTR(config-if)# desc # to MTL-RTR #
VAN-RTR(config-if)# ip add 3.3.3.1 255.255.255.252
VAN-RTR(config-if)# tunnel source g0/0/0
VAN-RTR(config-if)# tunnel destination 20.26.70.253
VAN-RTR(config-if)# tunnel mode gre ip
VAN-RTR(config-if)# no shut
VAN-RTR(config-if)# exit
```

Figure 4.8: Creating GRE tunnel interfaces using G0/0/0 as the source interface.

The final step to fully deploying the VPN is to configure tunnel interfaces. These tunnels tell the router where to establish the tunnels to and from which source. By setting the tunnel mode to GRE, the router activates the GRE encapsulation engine. This tells the interface to encapsulate any incoming packets (including multicast and broadcast messages) before handing it off to IPsec for encryption.

5) ENTERPRISE NETWORK SERVICES

5.1) Dynamic Host Configuration and Name Resolution

5.1.a) DHCPv4 Addressing

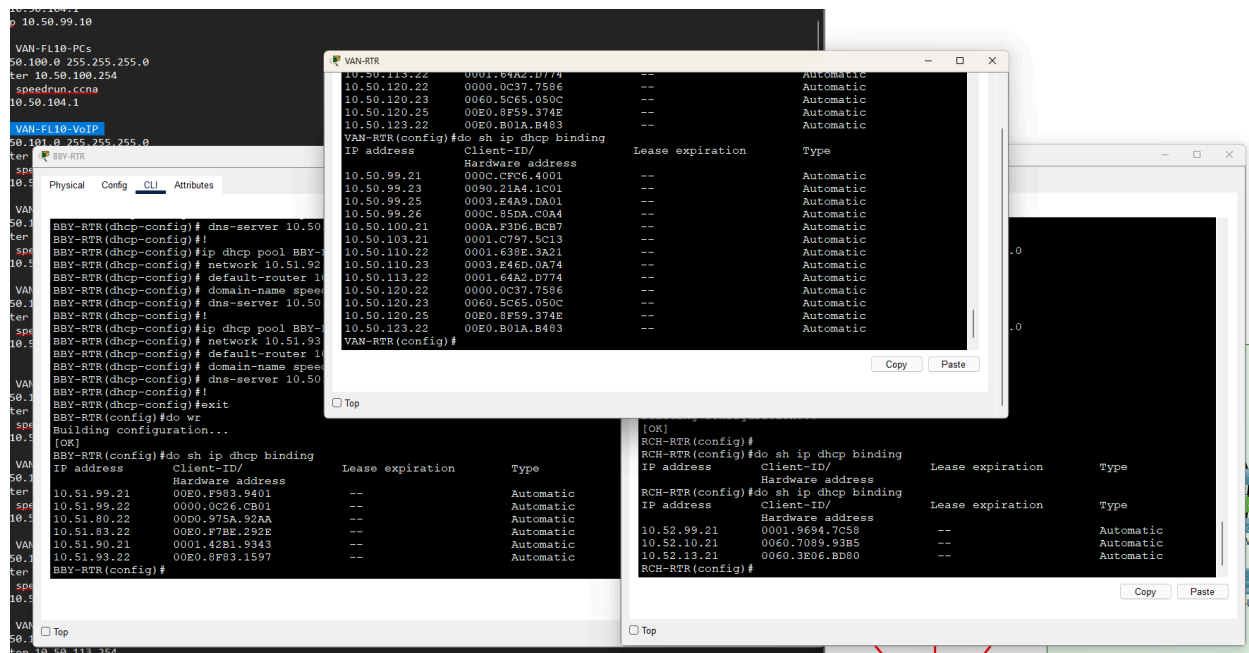


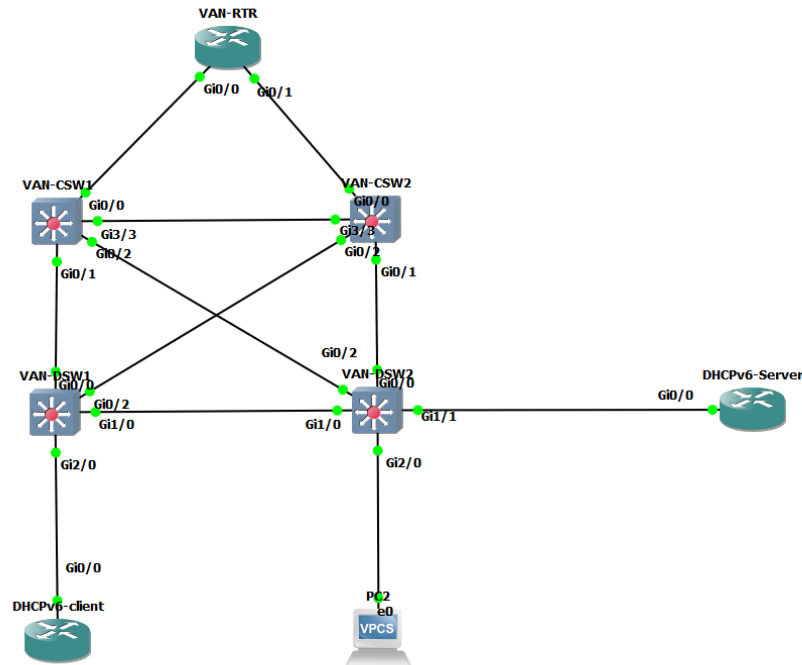
Figure 5.1: DHCPv4 binding tables on all routers

DHCPv4 is provided for all standard client VLANs, automating the assignment of IPv4 addresses, default gateway, domain name, and DNS server for all standard endpoints. **DHCP Option 43** will be provided to the Vancouver Wireless Access Points (APs) to automatically locate the **Wireless LAN Controller (WLC)**.

The DHCPv4 service is **fully decentralised** across the three networks. Each router will act as a local DHCP server to provide DHCP service exclusively to their LAN. For example, VAN-RTR will only provide the service to Vancouver VLANs, BBY-RTR for Burnaby VLANs, and RCH-RTR for Richmond VLANs. Distribution switches will then relay any DHCP message between the server and clients. .

The first 20 and last 14 usable addresses will be excluded from the pools to reserve static addresses for devices that need consistent identification. This would be the case for default gateways and most management devices.

5.1.b) Stateful DHCPv6 Addressing



```
Router(config)#
Router(config)#host DHCPv6-Client
DHCPv6-Client(config)#
DHCPv6-Client(config)#ipv6 uni
DHCPv6-Client(config)#ipv6 unicast-routing
DHCPv6-Client(config)#
DHCPv6-Client(config)#int g0/0
DHCPv6-Client(config-if)#
DHCPv6-Client(config-if)#ipv6 enable
DHCPv6-Client(config-if)#
DHCPv6-Client(config-if)#ipv6 add dhcp
DHCPv6-Client(config-if)#
DHCPv6-Client(config-if)#no sh
DHCPv6-Client(config-if)#
DHCPv6-Client(config-if)#
*Jun 21 22:03:16.924: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
DHCPv6-Client(config-if)#
*Jun 21 22:03:17.924: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
DHCPv6-Client(config-if)#
*Jun 21 22:03:21.712: %PNP-6-PNP DNS_SERVER_SET: DNS server (2001:10:50:14::101) set by (pid=308, pname=DHCPv6 client, time=22:03:21 UTC Sun Jun 21 2026)
*Jun 21 22:03:21.713: %PNP-6-PNP DOMAIN_NAME_SET: Domain name (speedrun.ccna) set on (GigabitEthernet0/0) by (pid=308, pname=DHCPv6 client, time=22:03:21 UTC Sun Jun 21 2026)
*Jun 21 22:03:21.751: %PNP-6-PNP DNS_SERVER_SET: DNS server (2001:10:50:14::101) set by (pid=308, pname=DHCPv6 client, time=22:03:21 UTC Sun Jun 21 2026)
DHCPv6-Client(config-if)#do sh ipv int b
DHCPv6-Client(config-if)#
GigabitEthernet0/0    [up/up]
    FE80::EAF:7CFF:FEDE:0
    2001:10:50:10:E9A5:3EEF:CEBD:A37F
    2001:10:50:10:FD4F:ED07:51A6:4129
GigabitEthernet0/1    [administratively down/down]
    unassigned
GigabitEthernet0/2    [administratively down/down]
    unassigned
GigabitEthernet0/3    [administratively down/down]
    unassigned
DHCPv6-Client(config-if)#
DHCPv6-Client(config-if)#
```

Figure 5.2: Demo GNS3 lab showing a DHCPv6 client obtaining a stateful DHCPv6 address from a router configured as a DHCPv6 server.

Stateful DHCPv6 is provided for all standard client VLANs, automating the assignment of IPv6 addresses, domain name, and DNS server for all standard endpoints. A dedicated DHCPv6 server will be provided for Vancouver and Burnaby LANs to centralise IPv6 management and ensure consistent auditing.

5.1.c) Local DNS Server Configuration

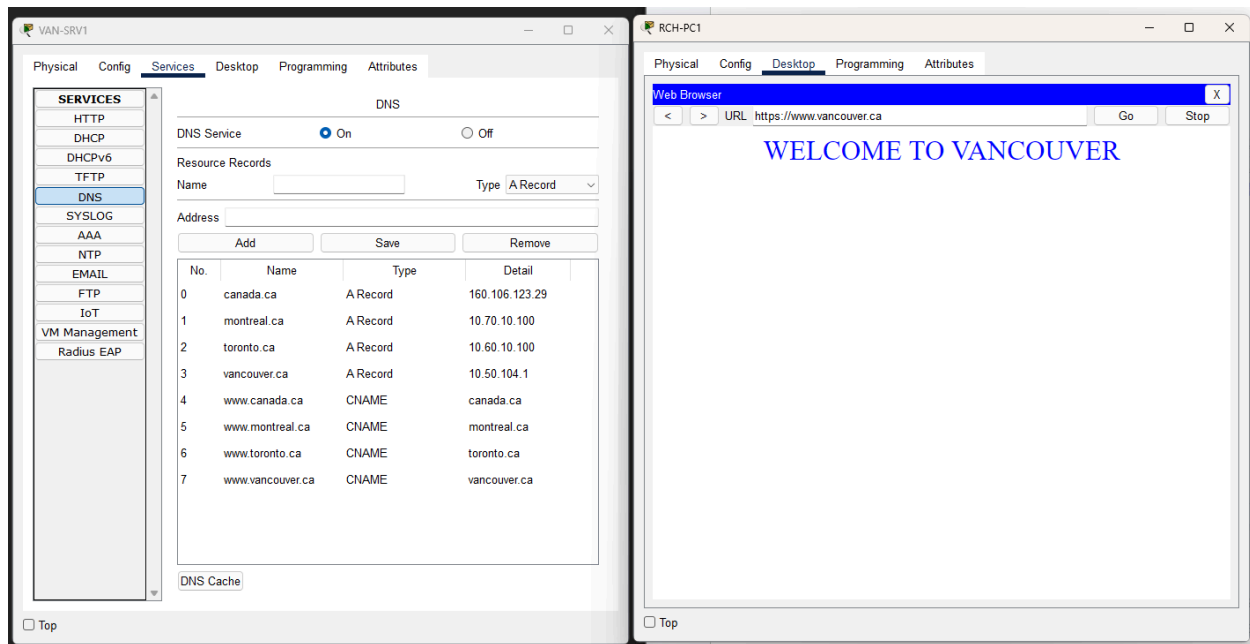


Figure 5.3: DNS records on VAN-SRV1

The local DNS service is provided in VAN-SRV1. It translates all of the **private corporate websites** (that of Vancouver, Toronto, and Montreal) into hostnames. This is to improve browsing speeds across the organisation and ensure Internet privacy by preventing external DNS servers from logging browsing habits.

5.2) Network Address Translation (NAT) Implementations

NAT will be implemented in all of Metro Vancouver, sharing the **20.26.50.0/24** block altogether to conserve public IP addresses. The IP block will be divided using **Variable Length Subnet Masking (VLSM)** to divide the Vancouver, Burnaby, and Richmond networks into multiple subnets of different sizes. This will minimise waste and maximise IP address utilisation relative to the size of the said networks.

5.2.a) Metro Vancouver NAT Pool Scheme

Office Network (Master subnet)	Public address pool	Public subnet	NAT mode
VANCOUVER HQ (20.26.50.0/25)	Floor10-PCs	20.26.50.0/27	Dynamic
	Floor11-PCs	20.26.50.32/27	Dynamic
	Floor12-PCs	20.26.50.64/27	Dynamic
	Floor10-WiFi	20.26.50.96/29	PAT
	Floor11-WiFi	20.26.50.104/29	PAT
	Floor12-WiFi	20.26.50.112/29	PAT
	VAN-SRV1	20.26.50.121/32	Static
BURNABY (20.26.50.128/26)	Floor8-PCs	20.26.50.128/28	Dynamic
	Floor9-PCs	20.26.50.144/28	Dynamic
	Floor8-WiFi	20.26.50.160/30	PAT
	Floor9-WiFi	20.26.50.164/30	PAT
RICHMOND (20.26.50.192/30)	PCs	20.26.50.193/32	PAT
	WiFi	20.26.50.194/32	PAT

Figure 5.4: Vancouver Public Address Pool Table

5.2.b) Static NAT, Dynamic NAT, and Port Address Translation (PAT)

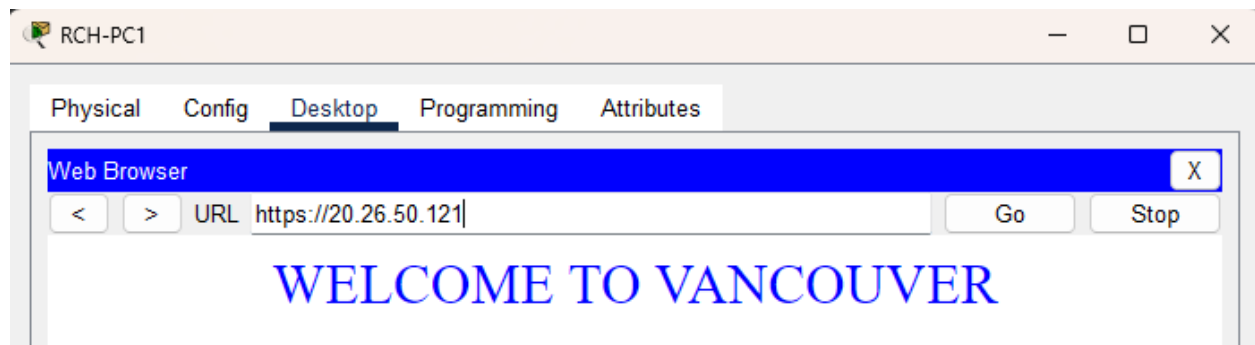


Figure 5.5: Browsing Vancouver's web server using its public IP address

Only VAN-SRV1 will use **static NAT** to provide a permanent one-to-one mapping between the private and public addresses of the server. The consistent reachability is essential for hosting public-facing services such as the HTTP/HTTPS web service.

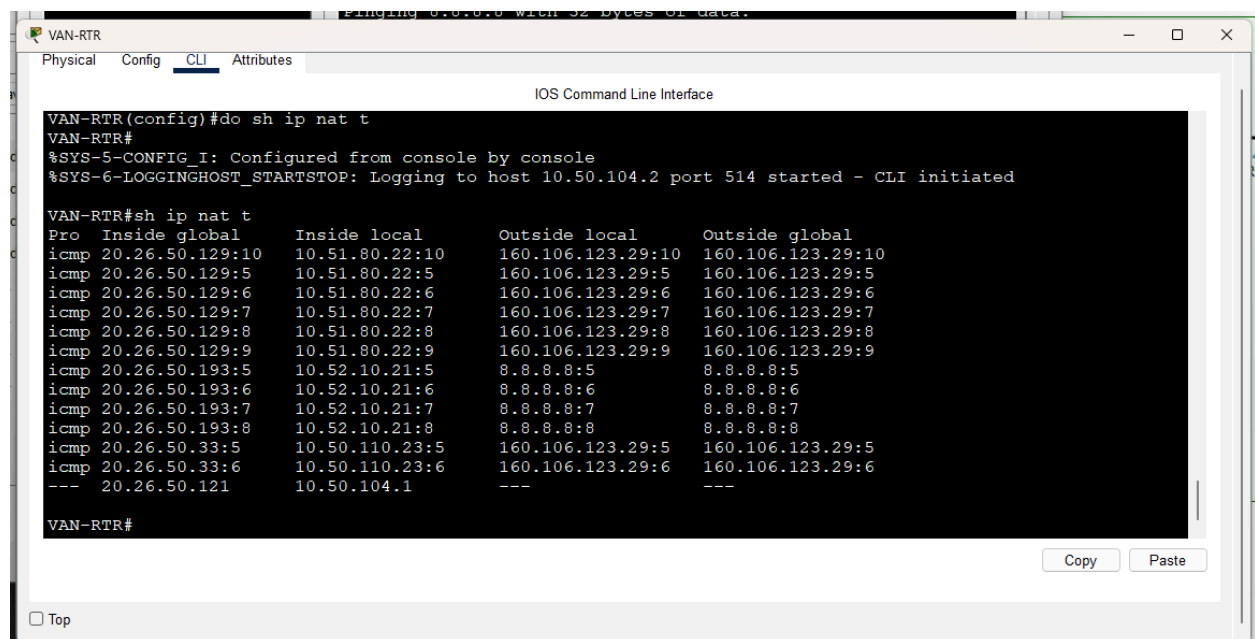


Figure 5.6: NAT table showing various NAT translations

To address the heavy bandwidth usage of the workstations (PCs), Vancouver and Burnaby will dynamically assign them a single public address per host. This will provide secure and direct one-to-one internet access without the network bottlenecking from NAT Overload (PAT).

Wireless LANs (WLANs) will use **port address translation (PAT)**. This allows for address conversation while theoretically allowing thousands of wireless clients within a small pool of IP

addresses. To further conserve the 20.26.50.0/24 block even further, all Richmond workstations and Wi-Fi devices will also use PAT. This is possible due to Richmond's small office size.

IP phones and network printers are excluded from public address mappings. They do not need Internet access to function properly, and doing so will pose security vulnerabilities.

5.3 Infrastructure Synchronisation (NTP)

All Cisco network devices will use NTP to synchronise their internal clocks to a highly accurate time source. This is vital for computer systems that depend on synchronised time to communicate securely, log events accurately, and maintain data integrity.

5.4 Resource Monitoring and Management (SNMP)

SNMP monitors and manages connected network devices from a central location. Integrating SNMP into the network devices gives IT teams total visibility and real-time data so problems can be spotted and fixed immediately. SNMPv3 will be used to enforce authentication and encryption, ensuring only authorised users can access sensitive network data.

5.5 Centralised System Logging (Syslog)

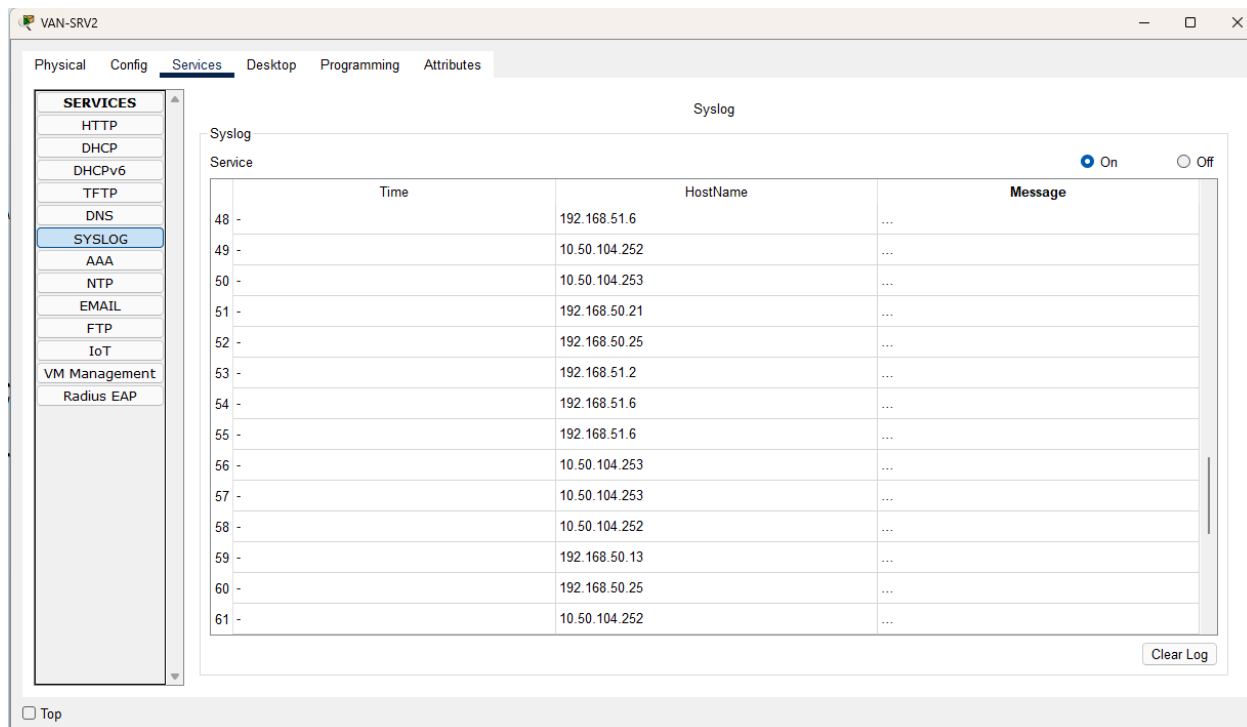


Figure 5.7: System logs on VAN-SRV2

All network devices across the enterprise Metro Vancouver network will log messages of all severity levels (0-7) to a centralized syslog server on VAN-SRV2. This provides a central location to filter for messages of any type and severity when an IT incident occurs.

6) WIRELESS ARCHITECTURE

6.1) Centralised Wireless Infrastructure

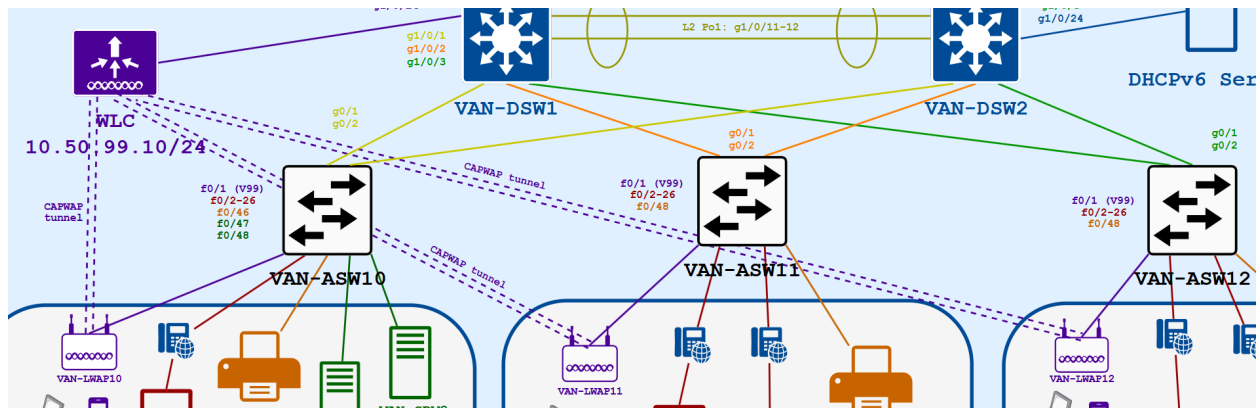


Figure 6.1: WLC deployment on Vancouver HQ. All Vancouver APs are lightweight, meaning they all rely entirely on the WLC to operate.

6.1.a) Unified Wireless LAN Controller Deployment

A wireless LAN controller (WLC) will be deployed in the Vancouver HQ's Management VLAN to meet Speedrunning's requirements for a centralised wireless infrastructure in their office. The WLC will be responsible for only the Vancouver APs. The Burnaby and Richmond APs will locally switch their own network traffic, eliminating the need to forward it to the WLC and waste bandwidth in the backbone routing network.

6.1.b) AP Provisioning (CAPWAP Tunnel Establishment)

CAPWAP tunnels are used by the WLC and Vancouver's lightweight APs to communicate. These tunnels allow for centralised wireless network management, allowing the controller to configure APs and encapsulate wireless client traffic across the wired network.

6.2) Wireless Security

6.2.a) Enterprise WPA3 for Advanced Encryption and Security

Wi-Fi Protected Access 3 (WPA3) Enterprise will be deployed for wireless security. It is the modern and most secure choice for wireless networks. It replaces pre-shared keys (PSK) with **Simultaneous Authentication of Equals (SAE)**, preventing attackers from capturing data and cracking passwords offline.

Management frames in WPA2 are **unencrypted**. WPA3 strictly requires **Protected Management Frames (PMF)** to encrypt and authenticate management frames, preventing de-authentication and spoofing attacks.

WPA3 supports **Perfect Forward Secrecy (PFS)**. It works by assigning unique encryption keys every session, ensuring that compromised session keys cannot be used to decrypt past or future network traffic.

6.2.b) 802.1X Port-based Network Access Control

Wireless devices that need to connect to the network will have to authenticate through **802.1X**. **802.1X** provides **port-based network access control**, securing networks by authenticating clients, known as **supplicants**, through an **authentication server**. If permitted, the **authenticator** provides network access to the client.

7) DEPLOYMENT CHALLENGES

There will be several challenges when configuring and maintaining a network of this size and scale. Here are some of the possible challenges:

- **Legacy IOS versions:** This is particular to the Catalyst 2960, where several security features are not supported by its current IOS such as Type 9 hashing and modern RADIUS commands.
- **Per VLAN Spanning Tree:** PVST+ can overwhelm the distribution switches, especially for a multi-VLAN deployment consisting of more than 10 VLANs. This can be mitigated by **Multiple STP (MSTP)** in a future deployment upgrade.
- **Outbound Traffic Bottlenecking:** Vancouver's core router is the only Internet-facing router within Metro Vancouver. All Internet traffic from all campuses will traverse this router, which can cause massive resource overhead.
- **Lack of VPN Scalability:** The VPN tunnel mesh uses static routes to each other's private networks. The lack of a dynamic routing protocol can make IP address changes extremely disruptive.
- **DHCPv6 Server Configuration:** Vancouver and Burnaby use a centralised DHCPv6 server requiring Layer 3 relay agents in the distribution layer. Troubleshooting DHCPv6 may involve downtime, and there are no redundancy systems to keep DHCPv6 fault tolerant.
- **NAT Pool Exhaustion:** Because the NAT pools for Metro Vancouver are limited, exhaustion of public addresses can be possible. This is mitigated by using PAT on wireless clients while providing dynamic 1:1 NAT on secured and authorised workstations.
- **NTP Synchronisation:** Cisco routers are known to be very slow with NTP synchronisation. This can cause delays or inaccuracies with syslog messages.
- **Lack of WLC Redundancy:** WLC is connected to VAN-DSW1. Any outage in VAN-DSW1 will disconnect the CAPWAP tunnel between the Vancouver Lightweight APs and the WLC, causing downtime for wireless clients.
- **Device Incompatibility with WPA3 Enterprise:** Older smartphone models such as the iPhone XR and older, and most other wireless devices manufactured in **2018** or earlier do not fully support WPA3 Enterprise.